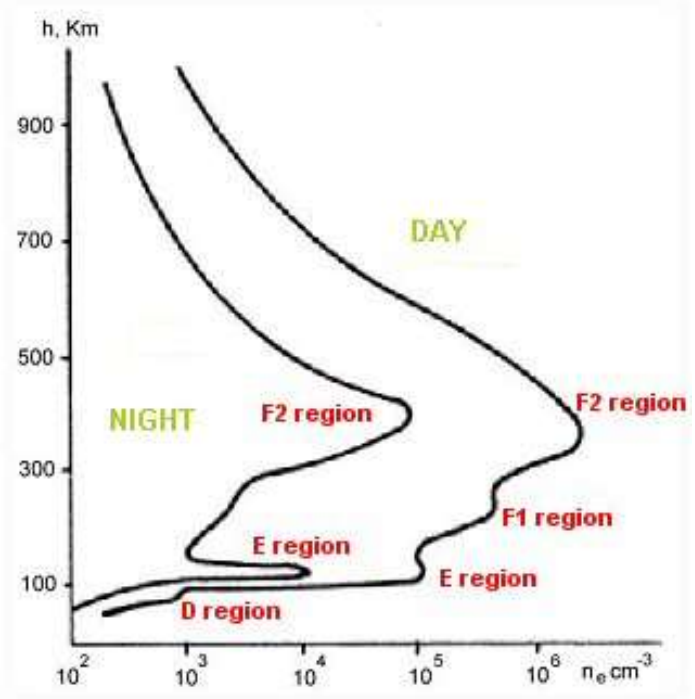
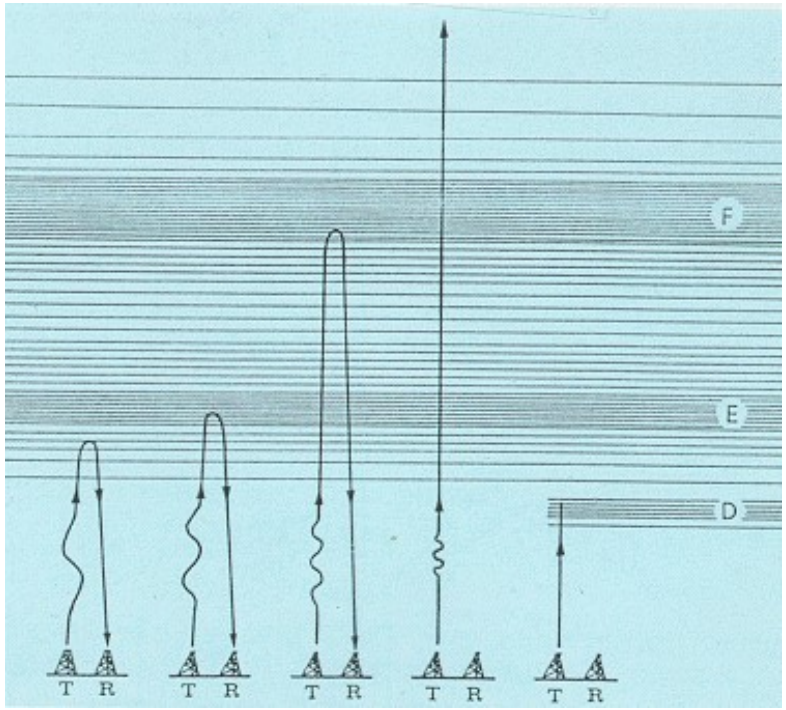
To the ionogram and after : a quick tour inside the AIS-INGV system

- Let's have a look to the principle which the measure is based on.
- Ionosphere is characterized by an electron density profile, that is the distribution of N along the height.

$$f_p = \sqrt{\frac{Nq^2}{4\pi^2 \varepsilon_0 m}} \approx 9\sqrt{N}$$

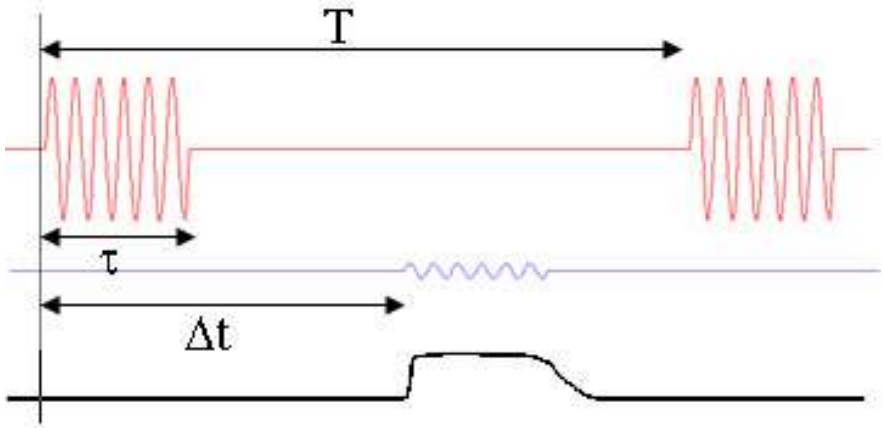




- For every used frequency, f, the reflection will happen at a height corresponding to the electron density able to zero the refraction index; particularly the reflection happens only if $f = f_p$.
- Since the plasma frequency depends on N varying with the height we have reflections from different altitudes.

Envelope radar

- Pulses of length equal to τ s repeating every T s are emitted trough a transmitting antenna.
- The power has to be adequate so that a detectable signal is obtained (design).



- Reflected and attenuated energy from the target reaches the RX antenna after an interval Δt.

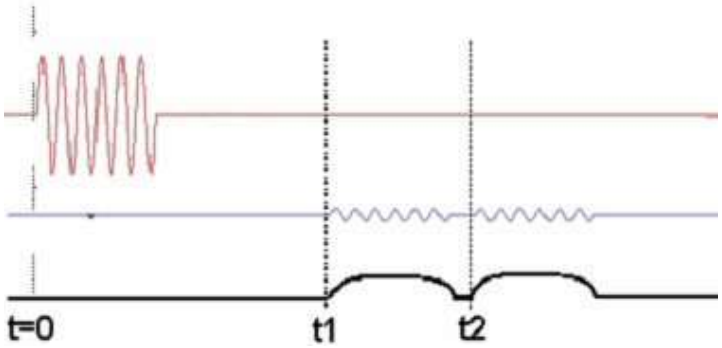
- A receiver tuned on the transmitted frequency generates a pulse whose length is approximately τ corresponding to the echo.

- According to this simple model we can derive the main features of an envelope radar:

- Target's distance ……

$$d = \frac{c \cdot \Delta t}{2}$$

- To evaluate a radar's resolution we need to remember that 2 echoes can be distinguished if the arriving times are so that $t_2-t_1>\tau$



- Minimum distance between 2 targets (resolution) …… $\delta d = \frac{c \cdot \tau}{2}$
- Energy from a P power amplifier…… $E = P \cdot \tau$
- Maximum target distance…… $d_{max} = \frac{c \cdot T}{2}$

Advantages

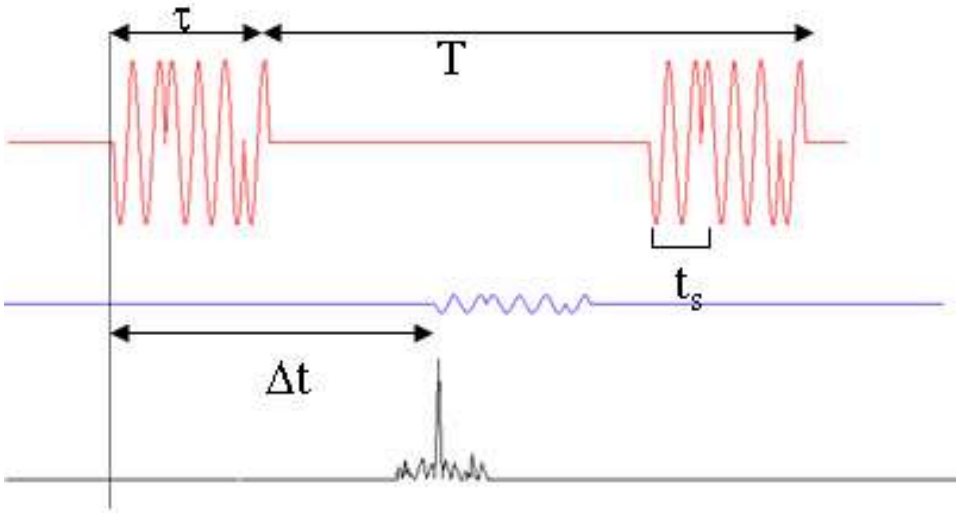
- Very simple TX system
- Very simple RX system
- There is the possibility of a complete analog receiver (no PC is needed)

Disadvantages

- Compromise for τ is needed
- Sometimes we have a limited resolution
- We have to use high power to get a good S/N

Pulse compression radar

- Varying repeatedly the phase inside the τ pulse we get the result that the resolution is the same corresponding to a narrower pulse called sub pulse t_s ("pulse compression").
- The definition of sub pulse is the minimum interval in which the phase is kept constant.
- The phase changes sequence is called CODE.



- The receiver is not completely analog; a mathematical process is necessary to look for the code in the received echo.
- The receiving process generates a very narrow peak corresponding to the delay (height) at which the code is found.
- It is a very robust process, with a very high noise immunity.

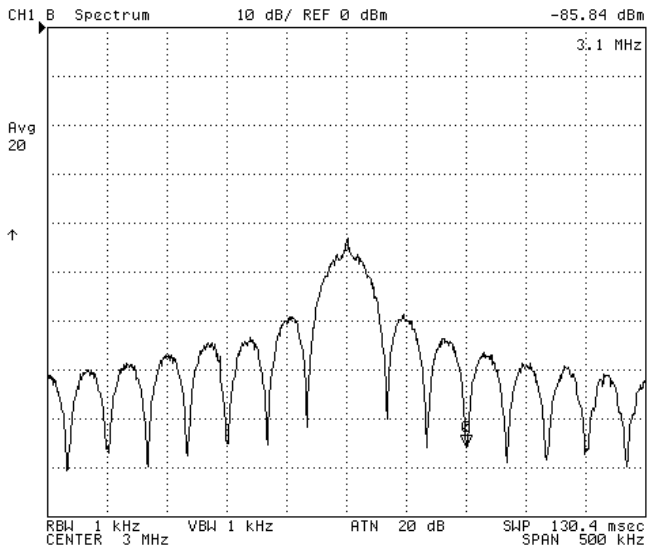
Advantages

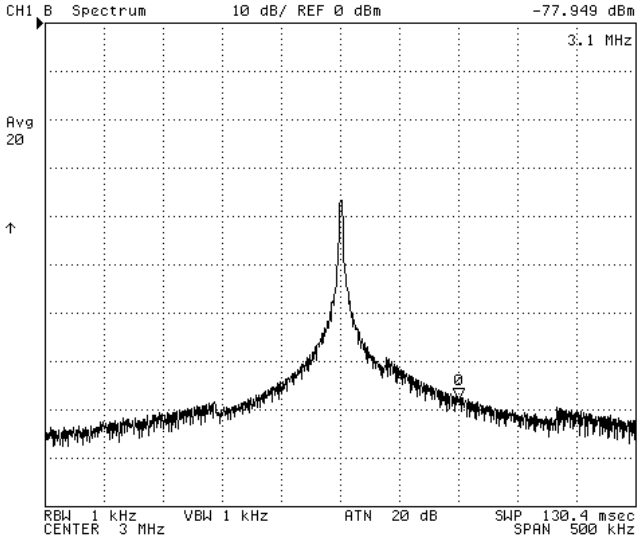
- Greater values for τ.
- Resolution depends on sub pulse length only.
- Less power to have the same energy.
- Higher S/N at receiver output.

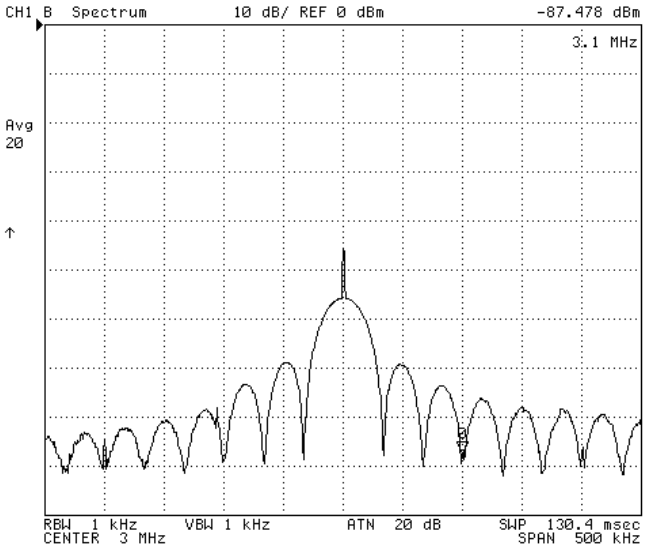
Disadvantages

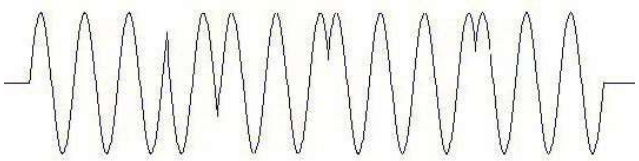
- More complexity in TX pulse generation.
- More complex analysis to detect the target.
- A mathematical process is necessary.

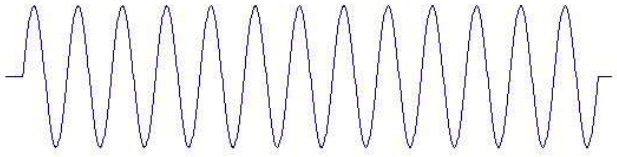
Why pulse compression works

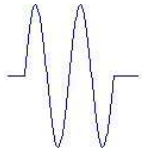












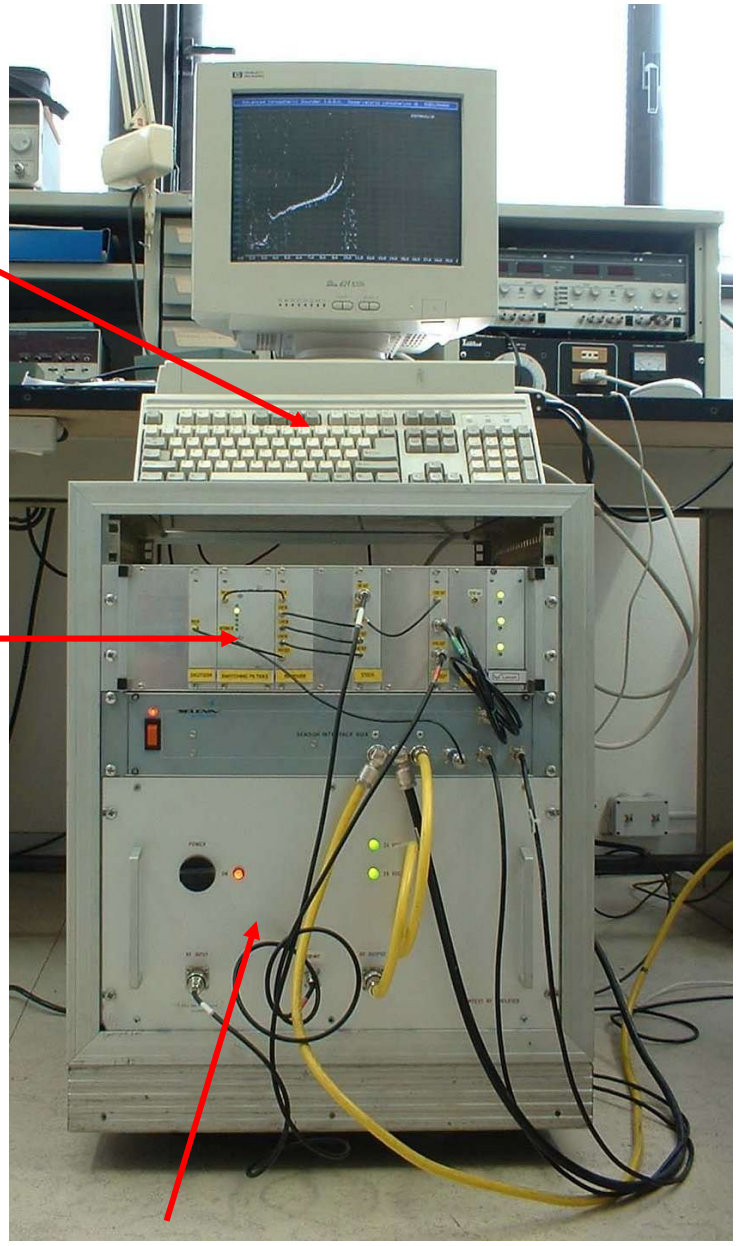
Pulse width	480µs
Repetition rate	60Hz
CODE	
Sub pulse	30µs
Band width	~66kHz

Pulse width	480µs
Repetition rate	60Hz
NO CODE	
Band width	~4kHz

Pulse width	30µs
Repetition rate	60Hz
NO CODE	
Band width	~66kHz

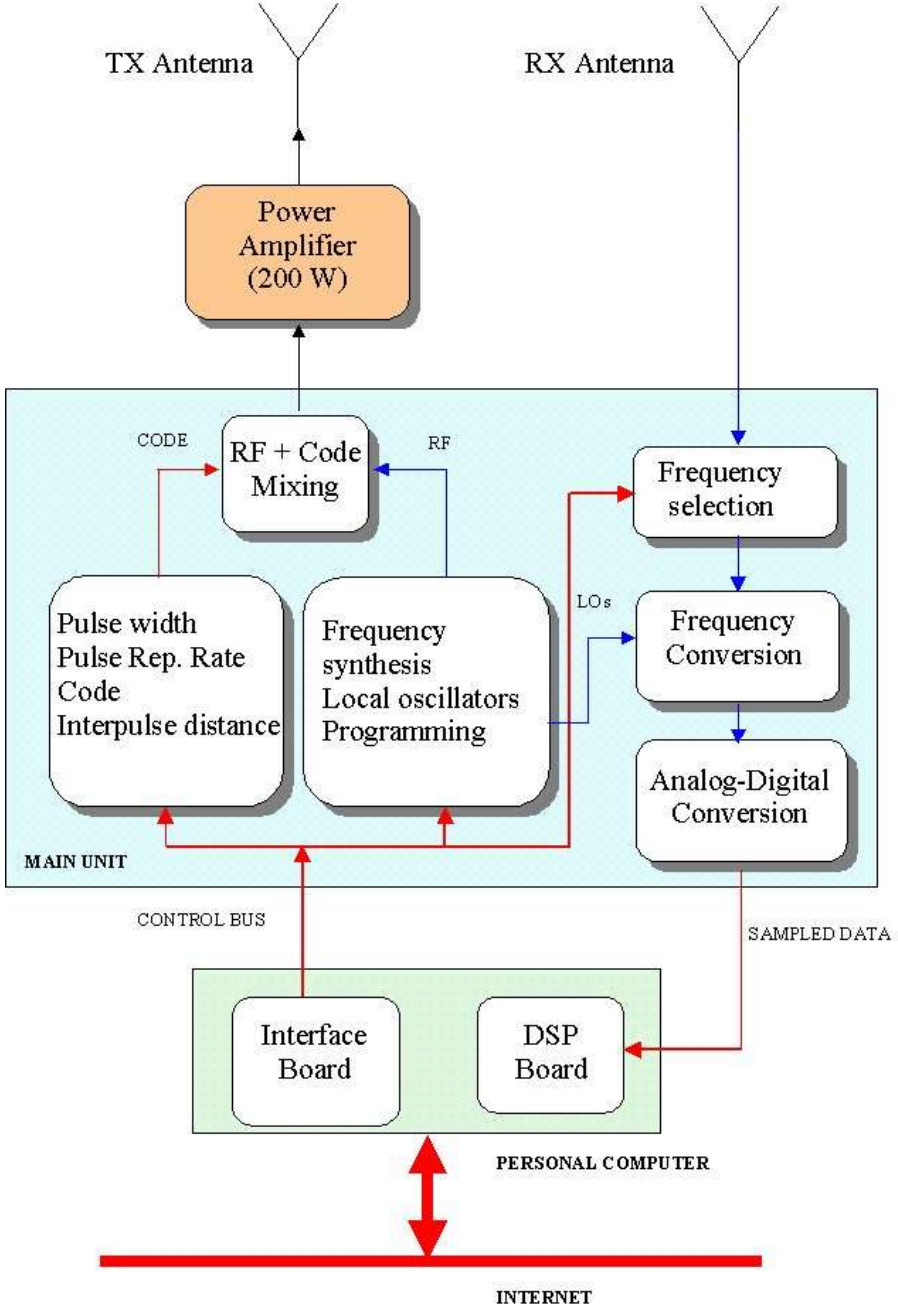
Advanced **I***onospheric* **S***ounder* AIS-INGV

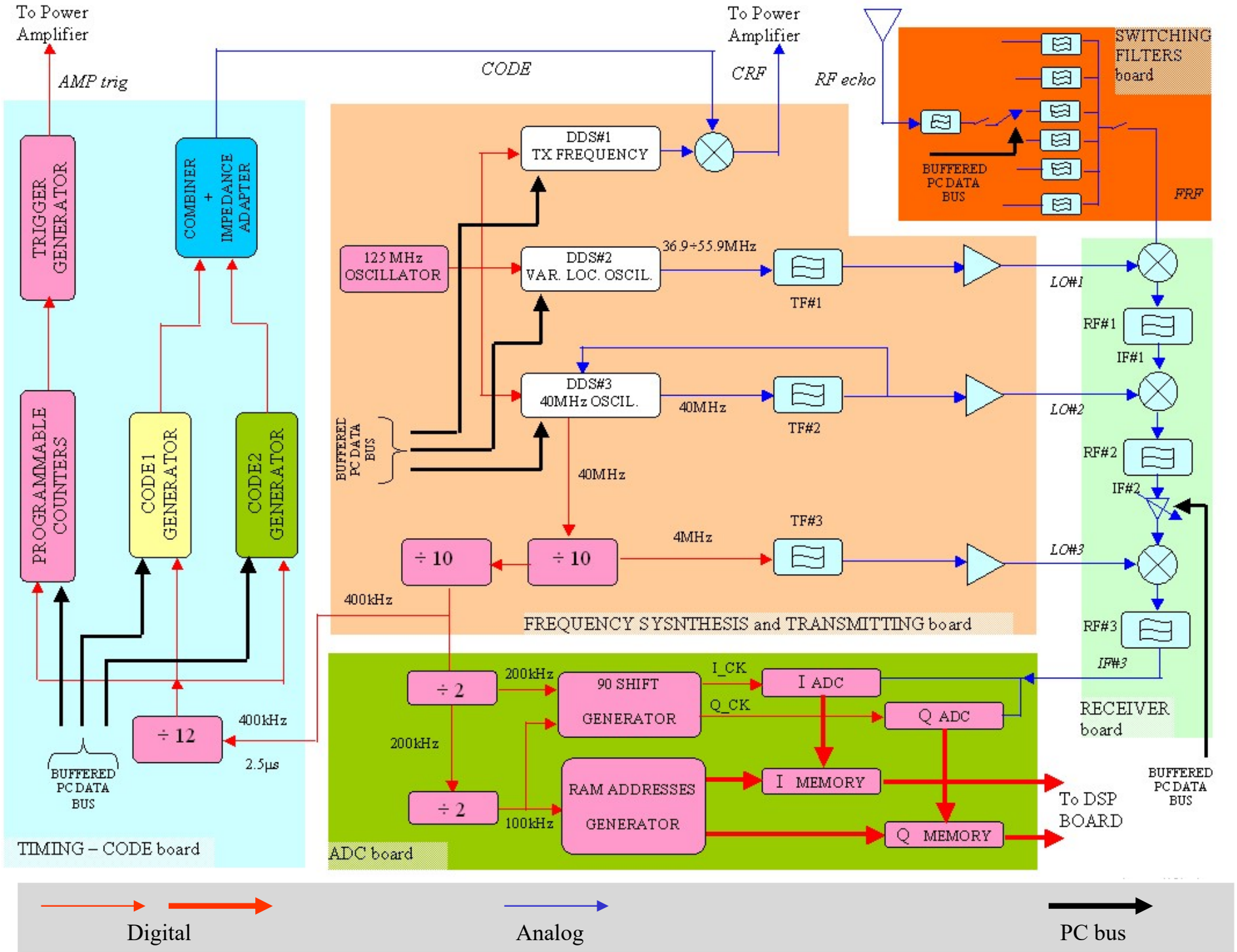
PC



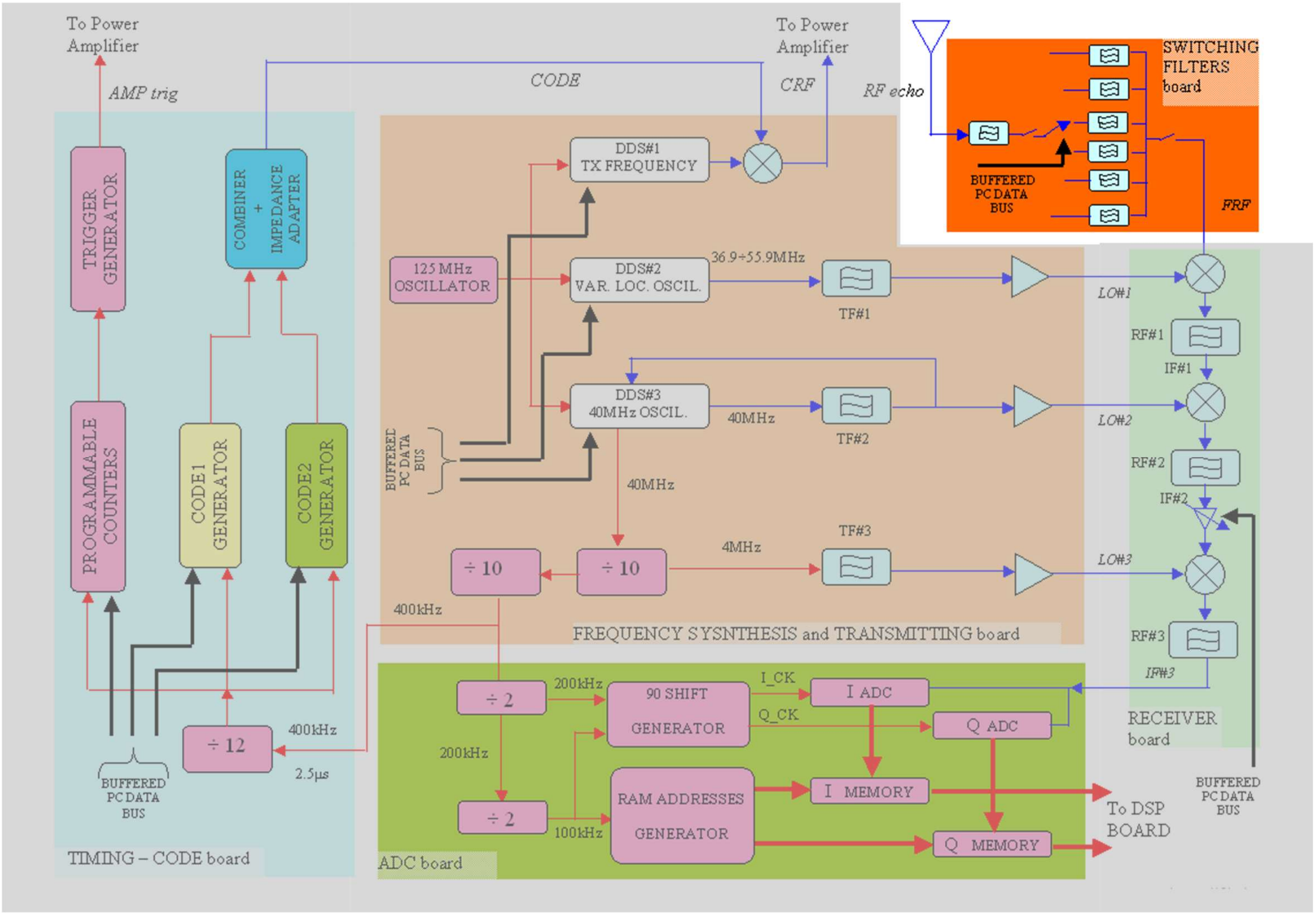
Main unit

Power amplifier



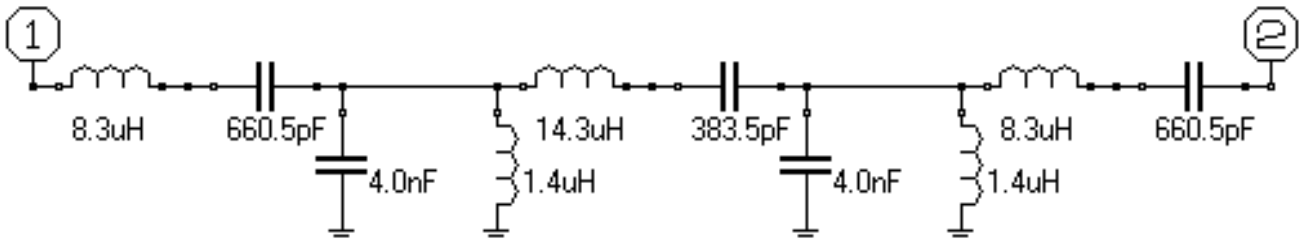

To Power Amplifier
AMP trig
CODE
To Power Amplifier
CRF
RF echo
SWITCHING FILTERS board
BUFFERED PC DATA BUS
FRF
TRIGGER GENERATOR
COMBINER + IMPEDANCE ADAPTER
DDS#1 TX FREQUENCY
125 MHz OSCILLATOR
DDS#2 VAR. LOC. OSCIL.
36.9÷55.9MHz
TF#1
LO#1
RF#1
IF#1
PROGRAMMABLE COUNTERS
CODE1 GENERATOR
CODE2 GENERATOR
BUFFERED PC DATA BUS
DDS#3 40MHz OSCIL.
40MHz
TF#2
LO#2
RF#2
IF#2
40MHz
TF#3
4MHz
÷ 10
÷ 10
LO#3
400kHz
FREQUENCY SYSNTHESIS and TRANSMITTING board
RF#3
IF#3
÷ 2
200kHz
90 SHIFT GENERATOR
I_CK
Q_CK
I ADC
Q ADC
RECEIVER board
200kHz
÷ 12
400kHz
2.5µs
BUFFERED PC DATA BUS
RAM ADDRESSES GENERATOR
I MEMORY
Q MEMORY
To DSP BOARD
BUFFERED PC DATA BUS
÷ 2
100kHz
TIMING – CODE board
ADC board
Digital
Analog
PC bus

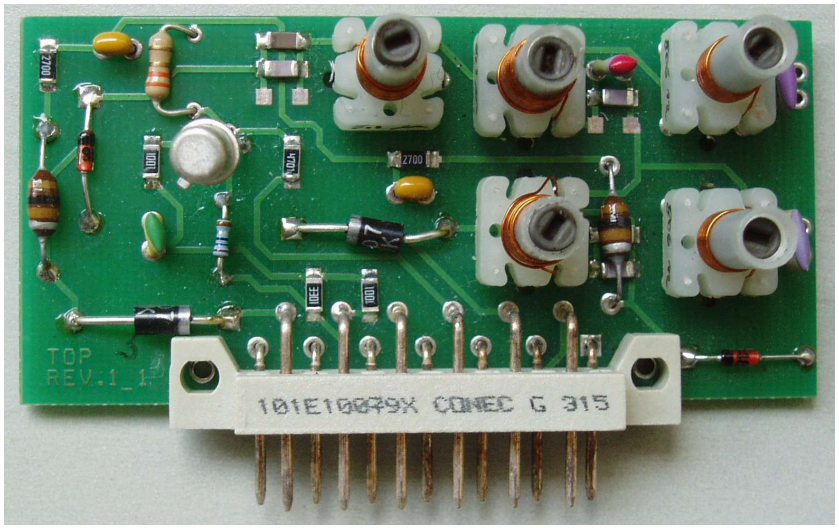
SWITCHING FILTERS


To Power Amplifier
AMP trig
CODE
To Power Amplifier
CRF
RF echo
SWITCHING FILTERS board
BUFFERED PC DATA BUS
FRF
TRIGGER GENERATOR
COMBINER + IMPEDANCE ADAPTER
DDS#1 TX FREQUENCY
125 MHz OSCILLATOR
DDS#2 VAR. LOC. OSCIL.
36.9÷55.9MHz
TF#1
LO#1
RF#1
IF#1
DDS#3 40MHz OSCIL.
40MHz
TF#2
LO#2
RF#2
IF#2
BUFFERED PC DATA BUS
40MHz
TF#3
4MHz
÷ 10
÷ 10
LO#3
RF#3
IF#3
400kHz
FREQUENCY SYSNTHESIS and TRANSMITTING board
PROGRAMMABLE COUNTERS
CODE1 GENERATOR
CODE2 GENERATOR
÷ 2
200kHz
90 SHIFT GENERATOR
I_CK
Q_CK
I ADC
Q ADC
RECEIVER board
200kHz
÷ 12
400kHz
2.5µs
BUFFERED PC DATA BUS
÷ 2
100kHz
RAM ADDRESSES GENERATOR
I MEMORY
Q MEMORY
To DSP BOARD
BUFFERED PC DATA BUS
TIMING – CODE board
ADC board

- Tasks: to reduce band for input frequencies (noise reduction)
  to isolate the receiver during the transmission phase.
- They are: 1 filter 1÷20MHz band active during listening time to limit the whole band;
  6 filters that are enabled or disable electronically during the sounding swept to reduce the band furtherly.
- Constant Q configuration has been chosen:

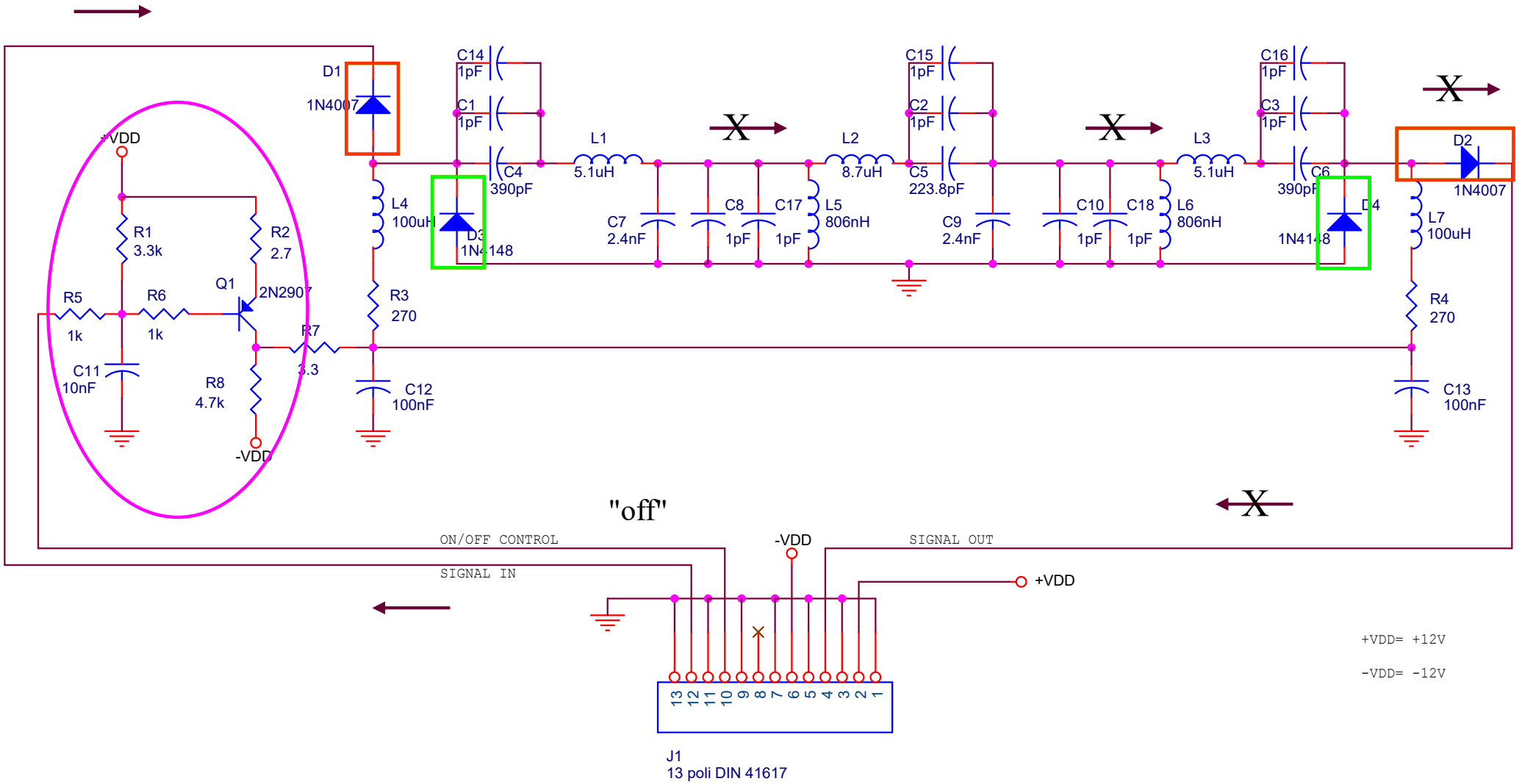
$$Q = \frac{B}{f_0} = k$$

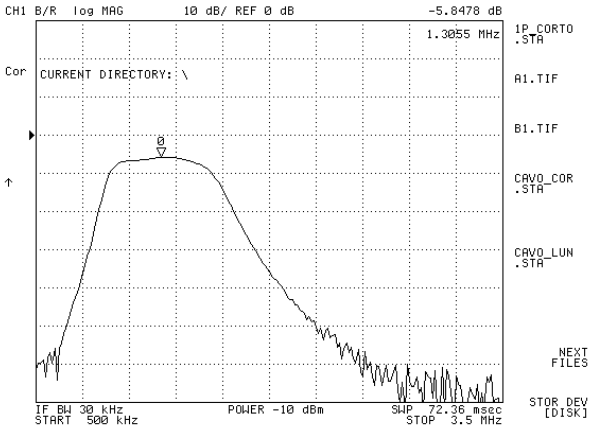




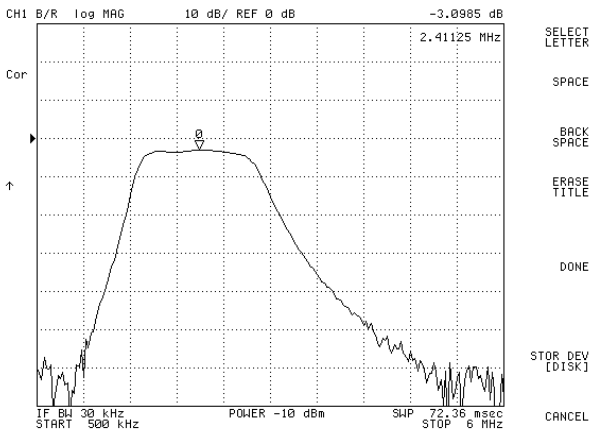
Central frequency: 2.2 MHz
Band: 1.1MHz

- They share common input and output lines, but they are not active contemporary.
- They are inserted or removed on the signal path thanks to a switching mechanism

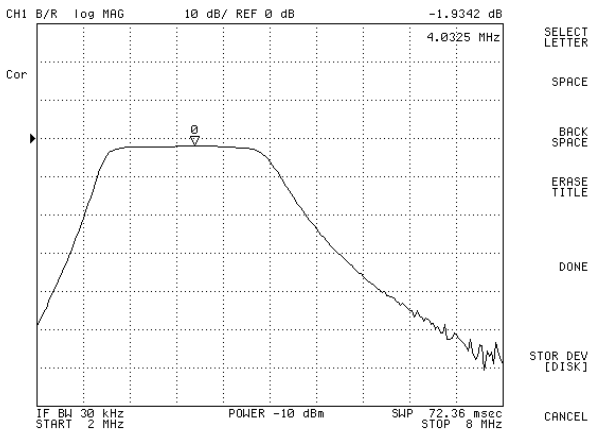

D1
1N4007
C14
1pF
C1
1pF
C4
390pF
L1
5.1uH
L2
8.7uH
C15
1pF
C2
1pF
C5
223.8pF
C16
1pF
C3
1pF
C6
390pF
D2
1N4007
+VDD
R1
3.3k
R2
2.7
Q1
2N2907
R5
1k
R6
1k
R7
3.3
R8
4.7k
C11
10nF
-VDD
L4
100uH
D3
1N4148
R3
270
C12
100nF
C7
2.4nF
C8
1pF
C17
1pF
L5
806nH
C9
2.4nF
C10
1pF
C18
1pF
L6
806nH
L3
5.1uH
D4
1N4148
L7
100uH
R4
270
C13
100nF
"off"
ON/OFF CONTROL
SIGNAL IN
-VDD
SIGNAL OUT
+VDD
13 12 11 10 9 8 7 6 5 4 3 2 1
J1
13 poli DIN 41617
+VDD= +12V
-VDD= -12V



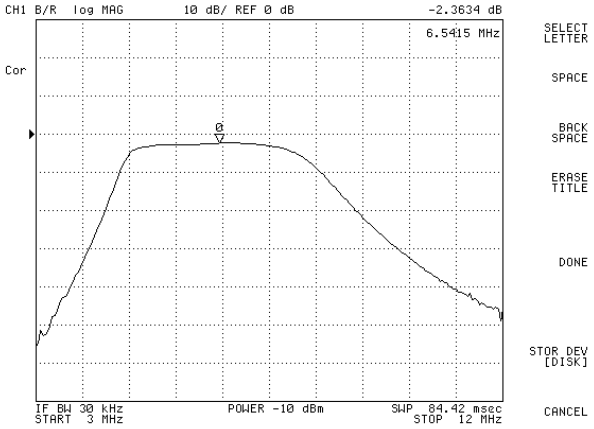
f_0=1.3MHz,
B=0.6MHz



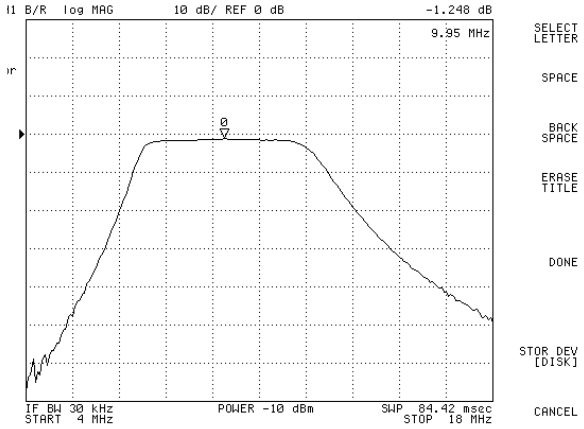
f_0=2.2MHz,
B=1.1MHz



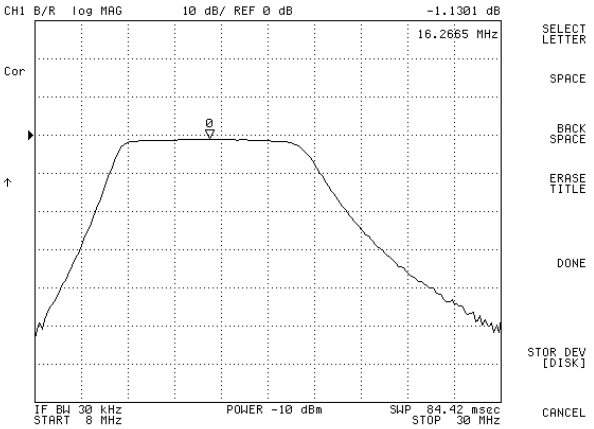
f_0=3.6MHz,
B=1.8MHz



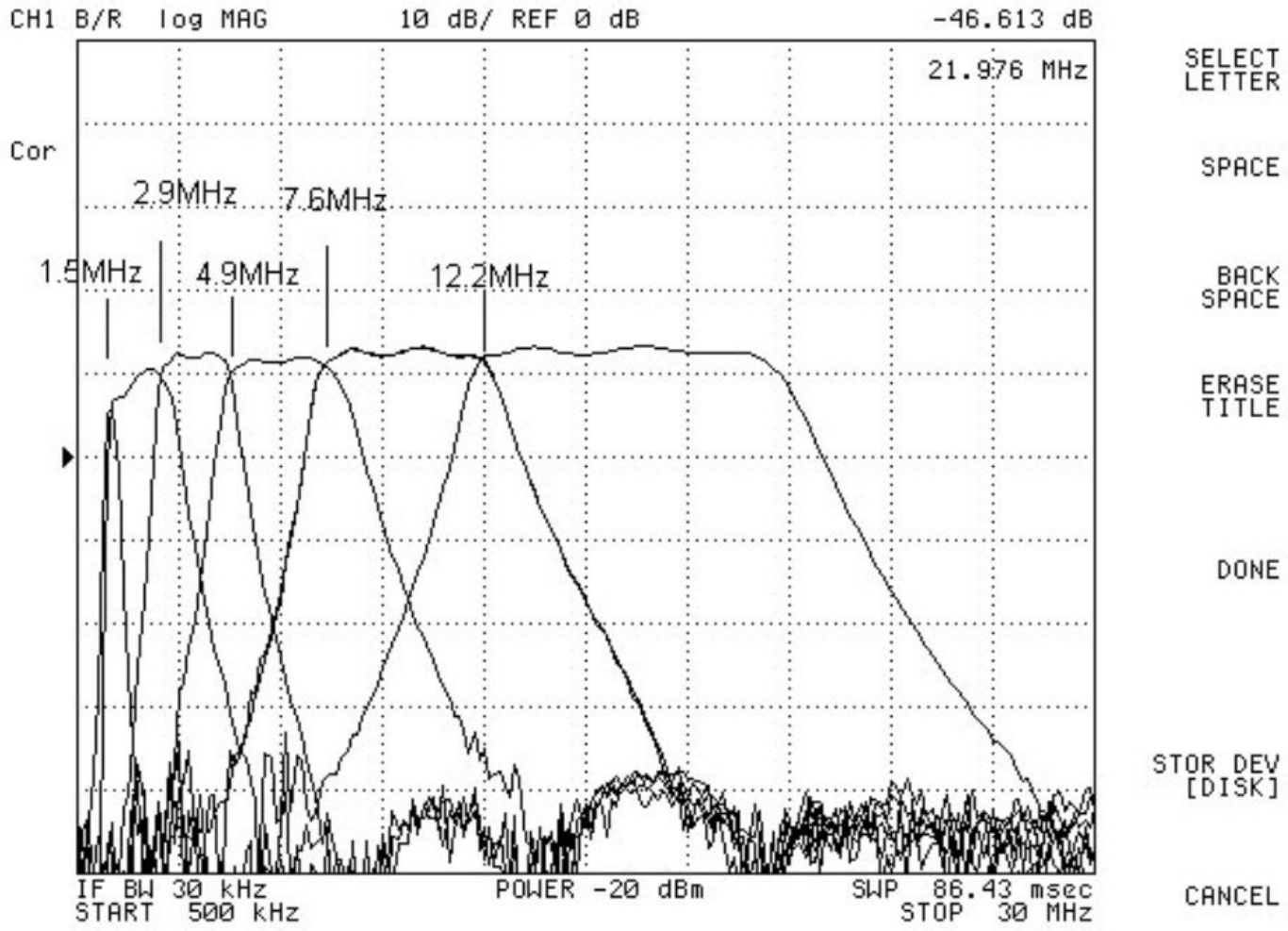
f_0=6.0MHz,
B=3.0MHz



f_0=10.0MHz,
B=5.0MHz

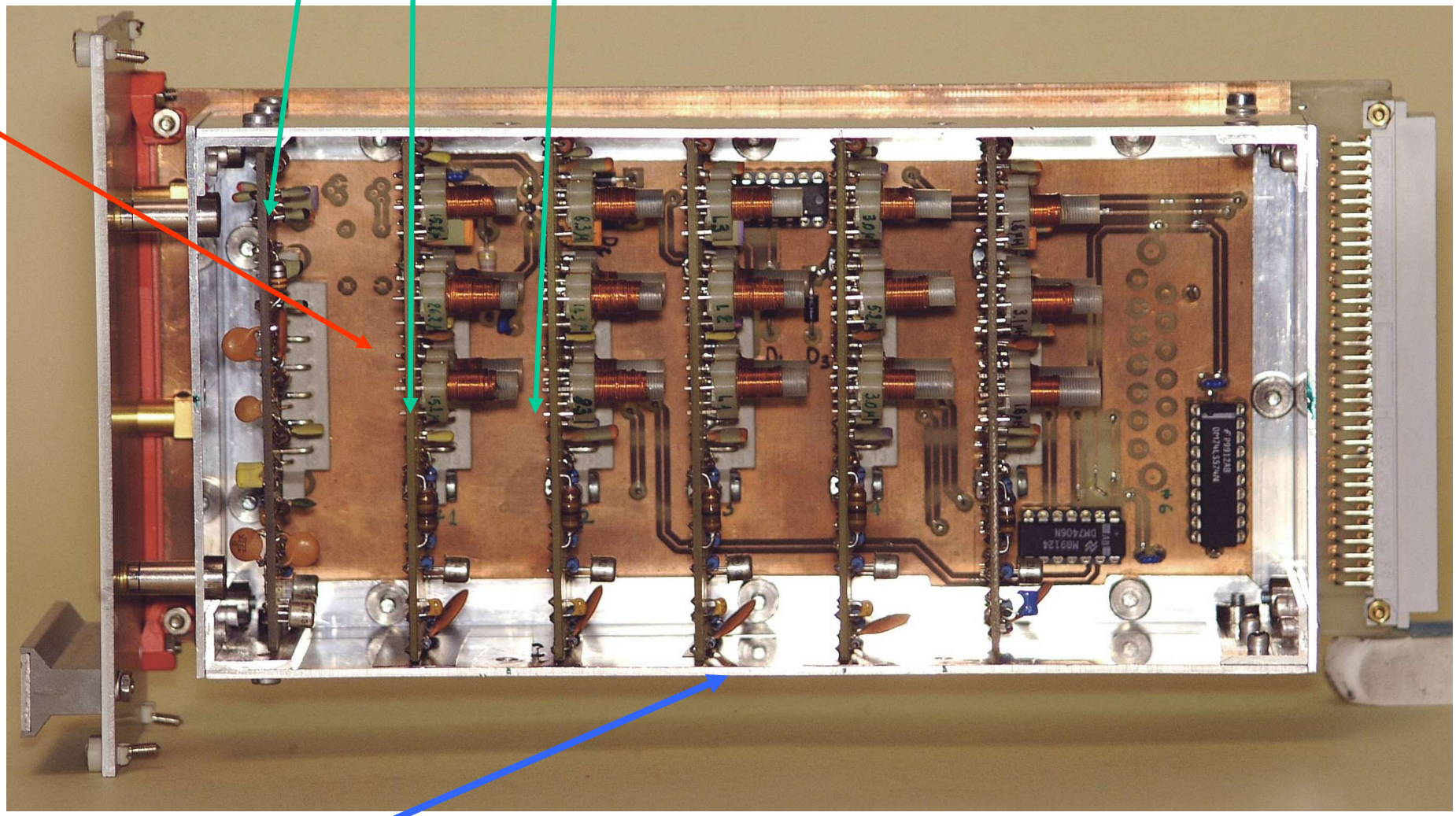


f_0=16.7MHz,
B=8.5MHz



- They cover the range from 1 to 20MHz
- Inequalities in gain are corrected with variable attenuator in the receiver

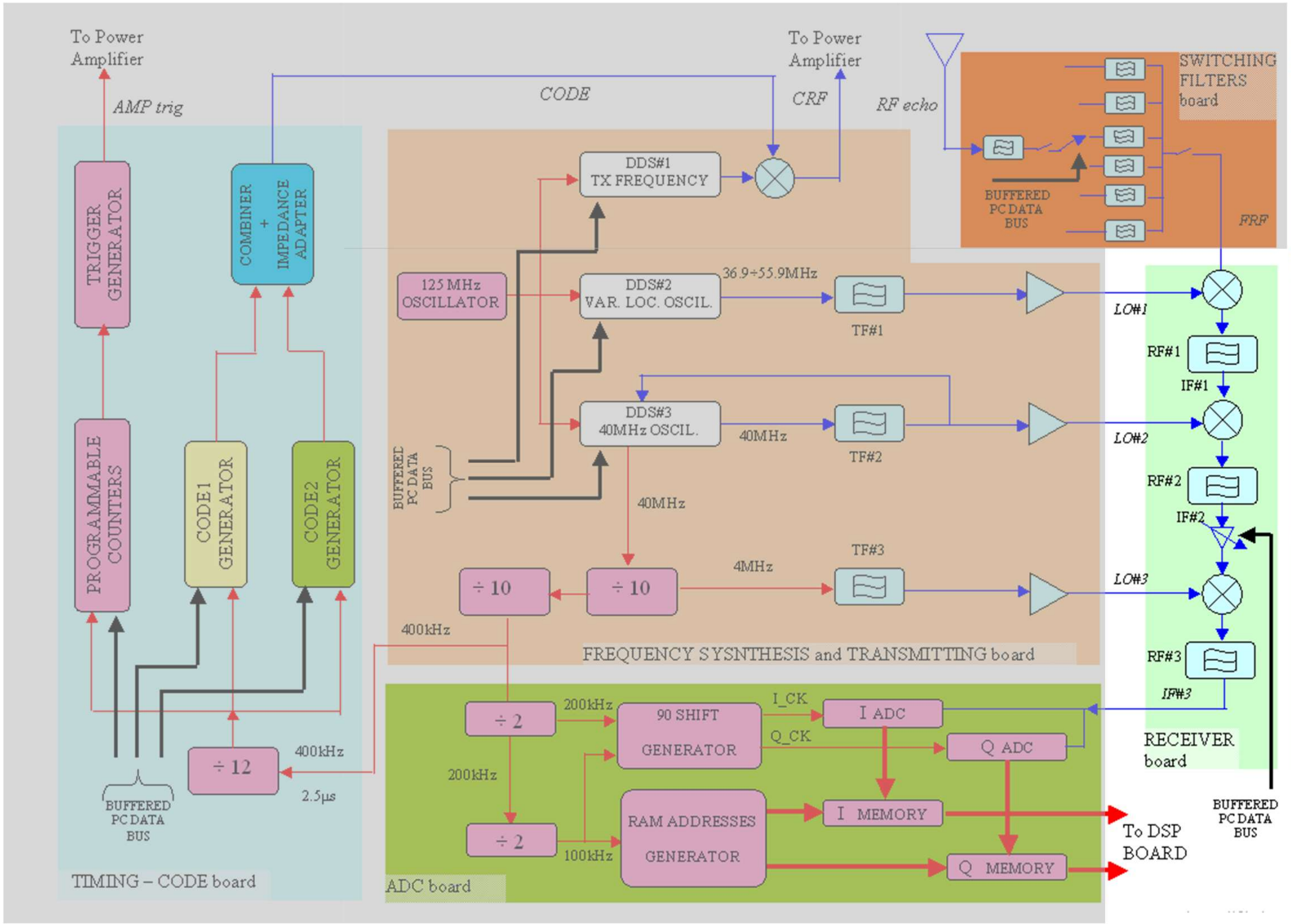
1÷20MHz #1 #2 #3 #4 #5 #6



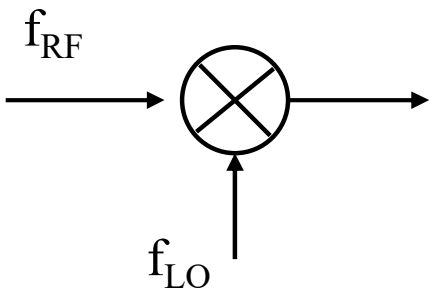
Ground plane

Shielding

RECEIVER


To Power Amplifier
AMP trig
CODE
To Power Amplifier
CRF
RF echo
SWITCHING FILTERS board
BUFFERED PC DATA BUS
FRF
TRIGGER GENERATOR
COMBINER + IMPEDANCE ADAPTER
DDS#1 TX FREQUENCY
125 MHz OSCILLATOR
DDS#2 VAR. LOC. OSCIL.
36.9÷55.9MHz
TF#1
LO#1
RF#1
IF#1
DDS#3 40MHz OSCIL.
40MHz
TF#2
LO#2
RF#2
IF#2
BUFFERED PC DATA BUS
40MHz
TF#3
4MHz
÷ 10
÷ 10
LO#3
RF#3
IF#3
400kHz
FREQUENCY SYSNTHESIS and TRANSMITTING board
PROGRAMMABLE COUNTERS
CODE1 GENERATOR
CODE2 GENERATOR
÷ 2
200kHz
90 SHIFT GENERATOR
I_CK
I ADC
Q_CK
Q ADC
RECEIVER board
200kHz
÷ 12
400kHz
2.5µs
BUFFERED PC DATA BUS
RAM ADDRESSES GENERATOR
I MEMORY
To DSP BOARD
BUFFERED PC DATA BUS
÷ 2
100kHz
Q MEMORY
TIMING – CODE board
ADC board

- Task: bring the information of the reflected signal from the ionosphere to a **unique** suitable **low** frequency (down conversion process).
- The frequency conversion is carried out by a mixer, a non linear device able to act as follows:



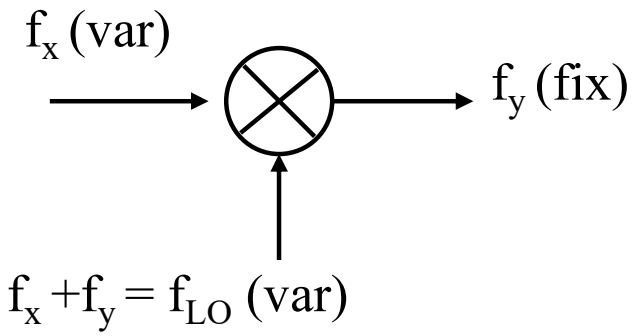
$$f_{IF} = |f_{RF} - f_{LO}|$$

and

$$f_{IF} = f_{RF} + f_{LO}$$

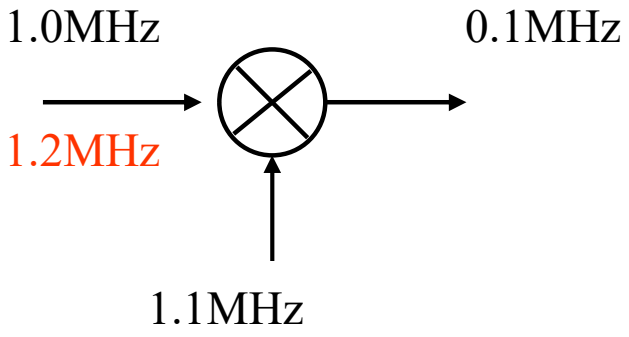
- Usually the output is filtered to get summation or difference of frequencies.

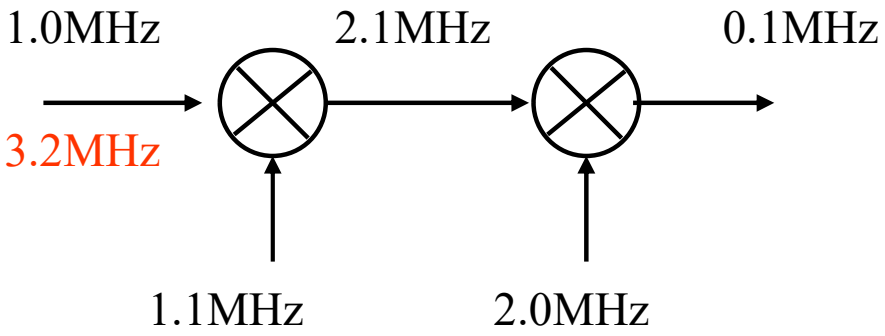
Super heterodyne receiver principle



How many conversions?

- One can think that going to the final low frequency directly could be a convenient way to design a receiver, but the following example reveals that things are different:



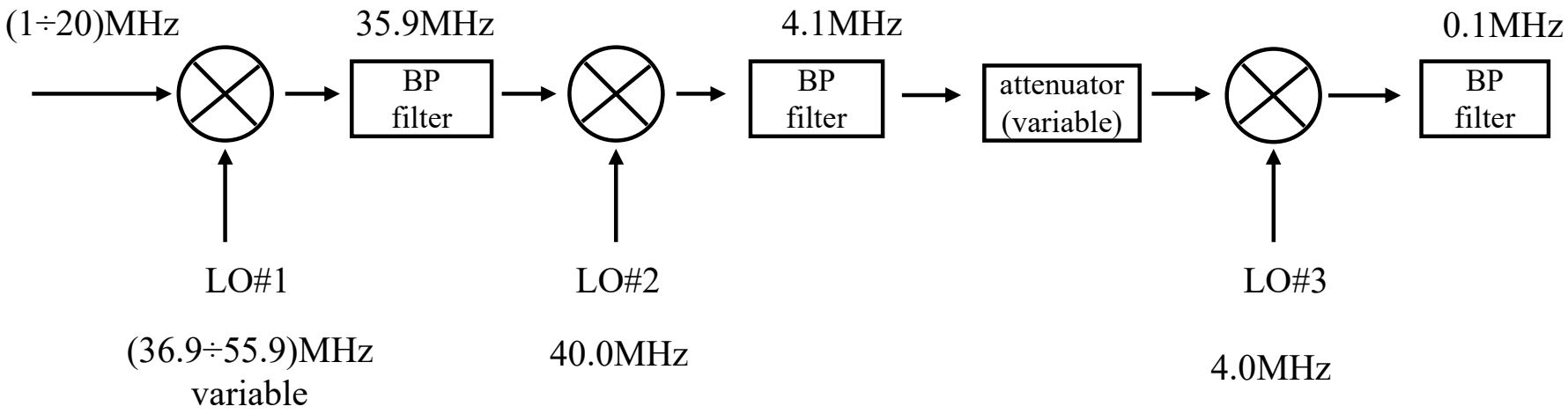


- Both black (1.0MHz, wanted) and red (1.2MHz unwanted) signals produce 0.1MHz beating with 1.1MHz LO, being very difficult to separate at the antenna level.

- Now black (1.0MHz, wanted) and red (3.2MHz unwanted) signals produce 2.1MHz beating with 1.1MHz LO, but now their separation is simple.

- The choice of down conversions number is an important point in a receiver design (highest is the number, less are the interference, but the complication increases).

Here the result!



- The success of pulse compression technique is based on the phase stability of the transmitted signals and the receiving process (remember that the code corresponds to phase shift).
- The only way to maintain phase locked is to generate all signals from a common clock.

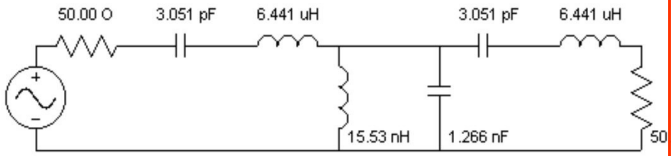
- A programmable attenuator is inserted in the receiving chain to vary the gain of the process (to equalize single switching filters, ….).

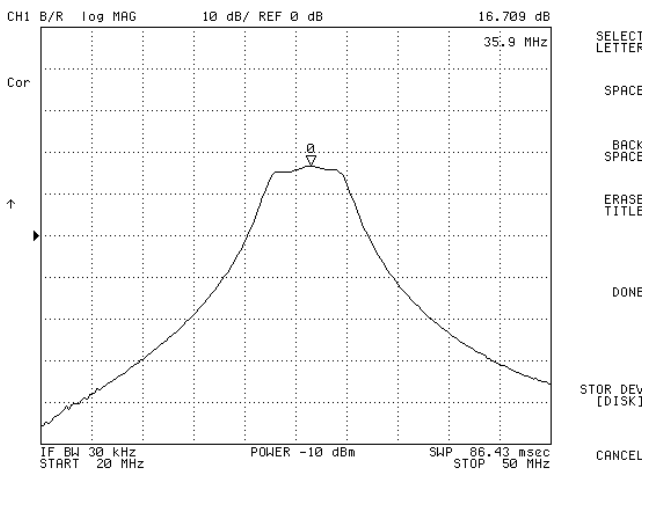
•Medium frequency values and corresponding filters (the information band is around 65kHz)

3°order chebyshev band pass

central freq. 35.9MHz

band 2.5MHz

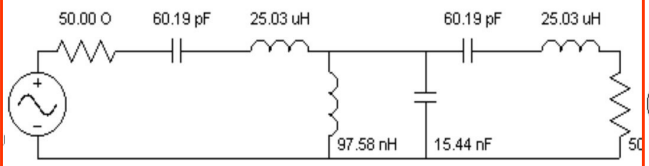


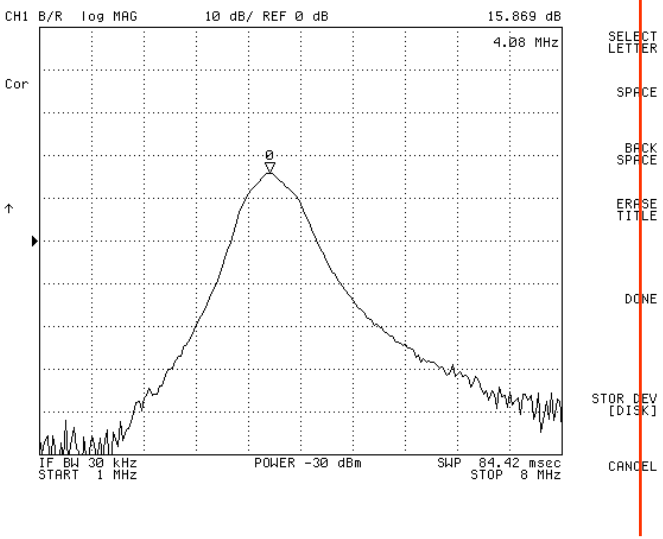


3°order chebyshev band pass

central freq. 4.1MHz

band 0.2MHz

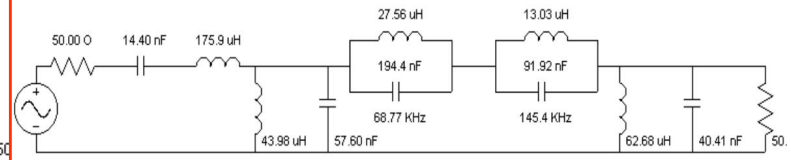


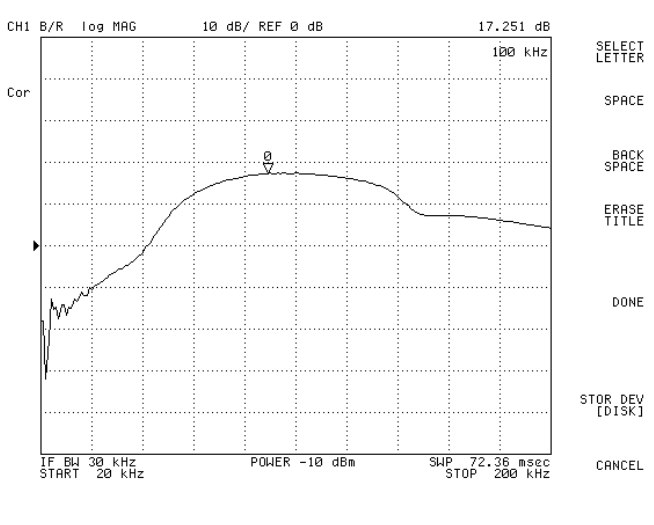


4° order elliptic band pass

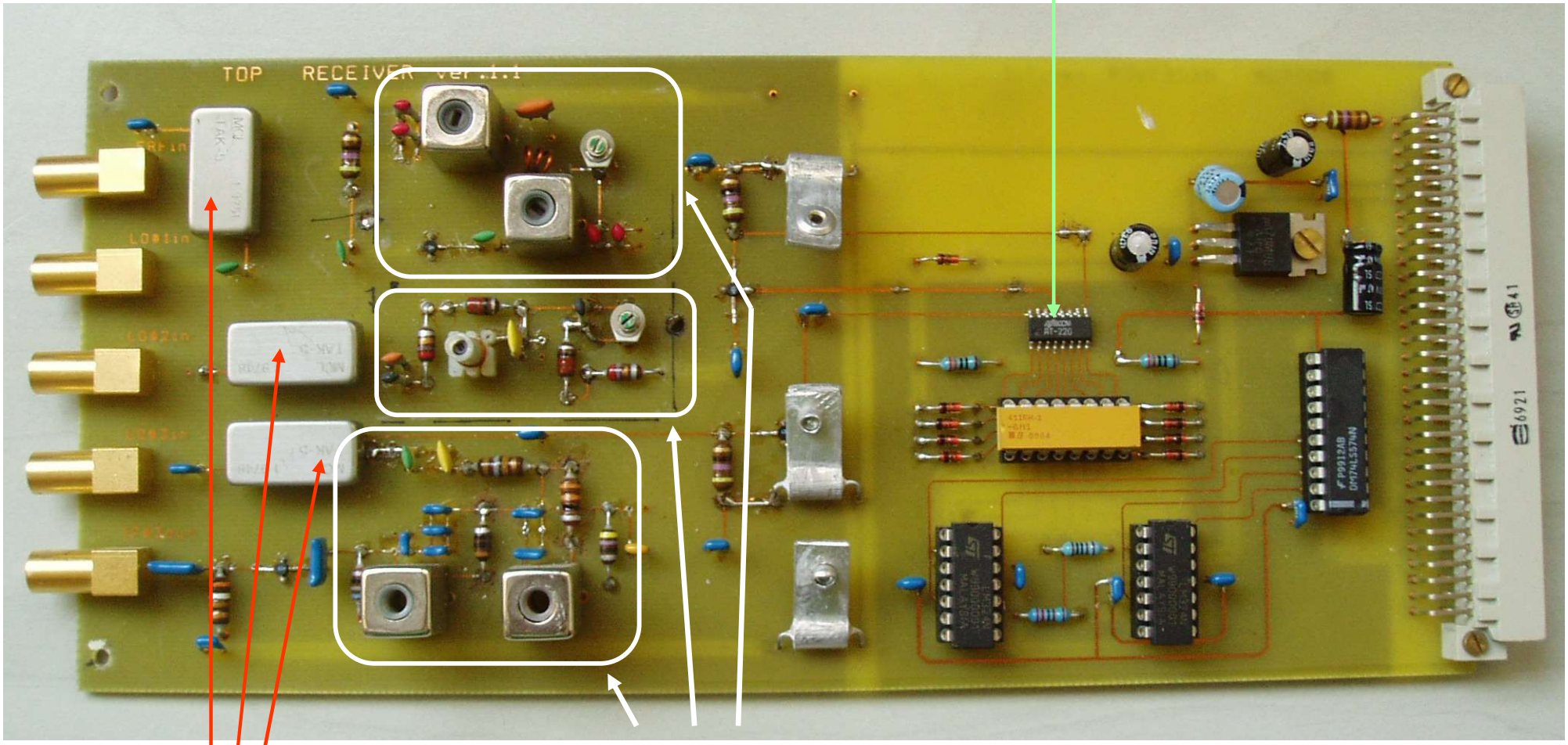
central freq. 0.1MHz

band 66kHz





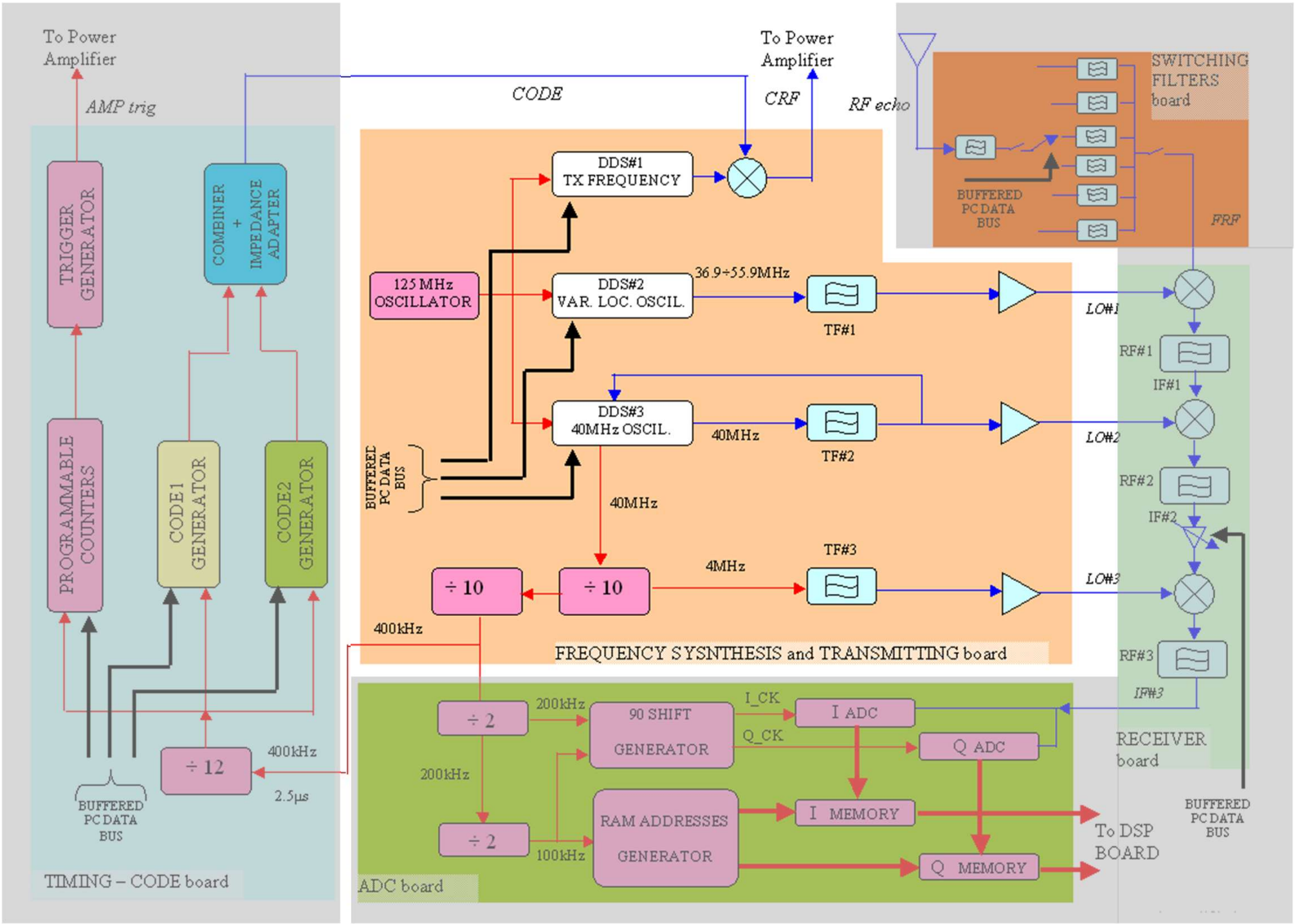
programmable
attenuator



filters

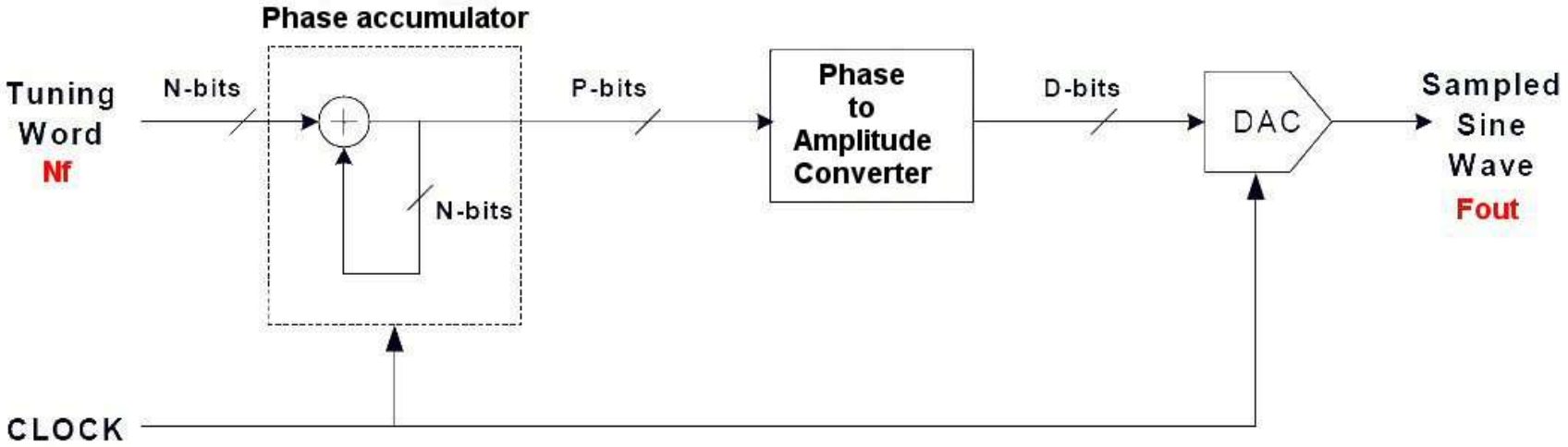
mixers

FREQUENCY SYNTHESIZER


To Power Amplifier
AMP trig
CODE
To Power Amplifier
CRF
RF echo
SWITCHING FILTERS board
BUFFERED PC DATA BUS
FRF
TRIGGER GENERATOR
COMBINER + IMPEDANCE ADAPTER
DDS#1 TX FREQUENCY
125 MHz OSCILLATOR
DDS#2 VAR. LOC. OSCIL.
36.9÷55.9MHz
TF#1
LO#1
RF#1
IF#1
PROGRAMMABLE COUNTERS
CODE1 GENERATOR
CODE2 GENERATOR
BUFFERED PC DATA BUS
DDS#3 40MHz OSCIL.
40MHz
TF#2
LO#2
RF#2
IF#2
40MHz
TF#3
4MHz
÷ 10
÷ 10
LO#3
400kHz
FREQUENCY SYSNTHESIS and TRANSMITTING board
RF#3
IF#3
÷ 2
200kHz
90 SHIFT GENERATOR
I_CK
I ADC
Q_CK
Q ADC
RECEIVER board
200kHz
÷ 2
100kHz
RAM ADDRESSES GENERATOR
I MEMORY
Q MEMORY
To DSP BOARD
BUFFERED PC DATA BUS
÷ 12
400kHz
2.5µs
BUFFERED PC DATA BUS
TIMING – CODE board
ADC board

DDS principles

- Frequency generation is based on the Direct Digital Synthesis (DDS) technique.
- DDS is a digital technique for generating a variable frequency sine wave from a fixed-frequency clock source.



Basic DDs block diagram

- For every clock pulse the phase is increased by N_f and an amplitude's value is generated
- The higher is N_f the shorter is the time to complete a period, the higher is the frequency of the sine wave

$$F_{out} = \frac{N_f \cdot CLOCK}{2^N}$$

where F_{OUT} = the output frequency of the DDS
N_f = the binary tuning word
CLOCK = the internal reference clock frequency (system clock)
N = The length in bits of the phase accumulator

DDS advantages

- FREQUENCY digitally tunable (typically with sub-Hertz resolution).
- PHASE is digitally adjustable, as well,
- NO ERRORS from drift due to temperature or aging of components.
- Two generators from the same clock will have phase locked frequency.

DDS restrictions

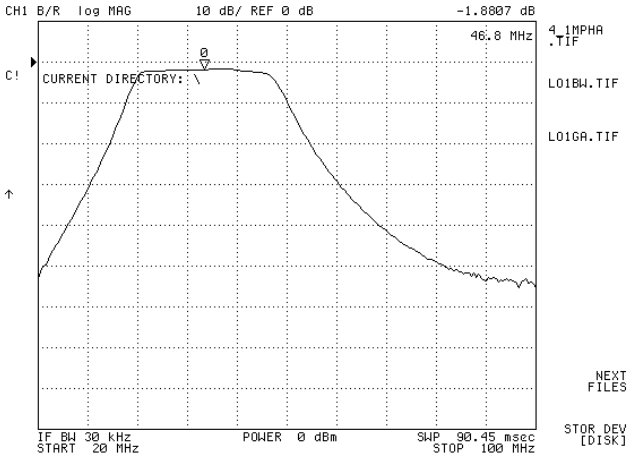
- FREQUENCY must be less than or equal to 1/2 the clock source frequency.
- The sine wave AMPLITUDE is fixed.
- Since the sine wave is digitally generated by using sampling techniques, there is a certain amount of DISTORTION. That is, the sine wave is not spectrally "pure", and filters must be used.

- Tasks: generates transmitting frequency (sine wave),
  creates the coded pulses,
  generates local oscillators for the receiver,
  generates the basic clock to have a phase locked process.

- Thanks to DDS devices we have a well known relationship between sine waves.
- It has 3 DDS sine wave generators with a common clock
  - DDS#1 generating transmitting frequency (1÷20)MHz
  - DDS#2 generating LO#1 accordingly with TX frequency (variable)
  - DDS#3 generating LO#2 (fix) and through a digital division process generates also LO#3 (fix)

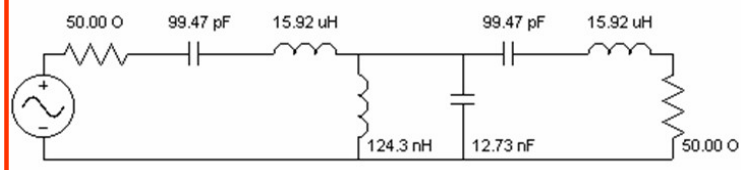
- The common clock has a frequency of 125MHz

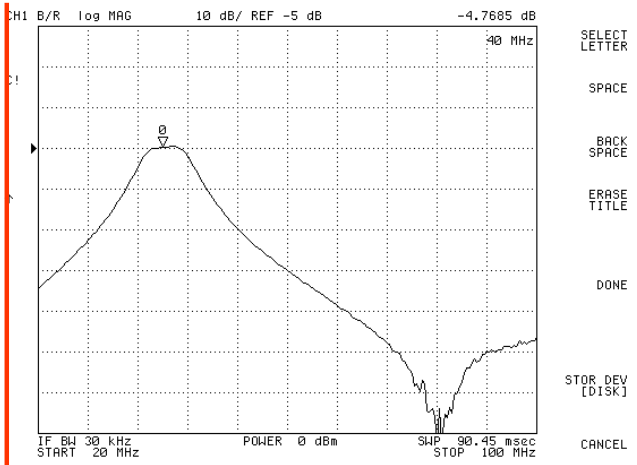
•Local oscillators are required to be as pure as possible, so filters are used



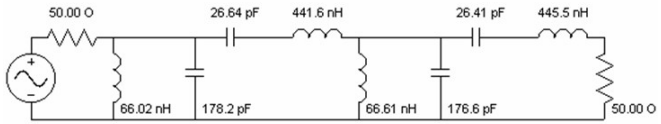
f_0=46.4MHz,
B=28.0MHz

• 3°order Butterworth band pass

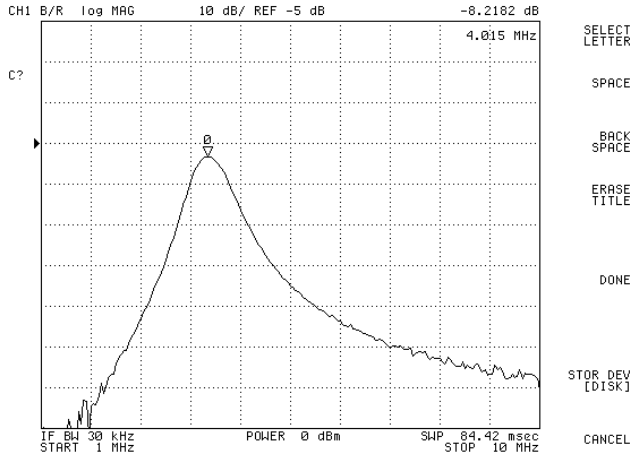




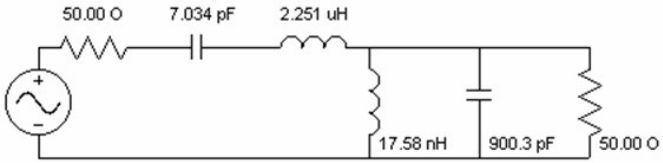
f_0=40.0MHz,
B=5.0MHz



• 4°order Chebychev band pass

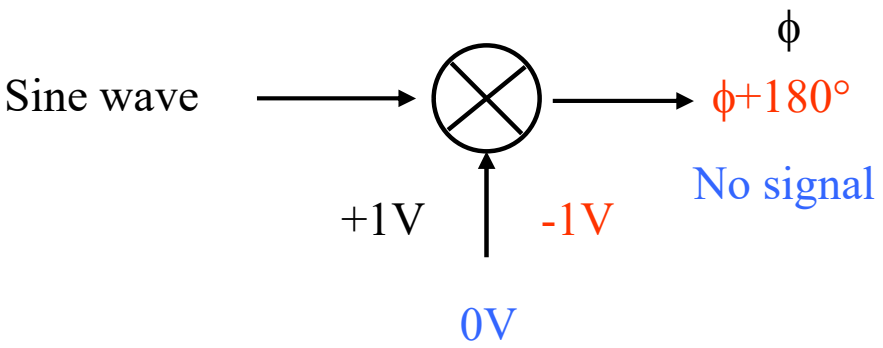


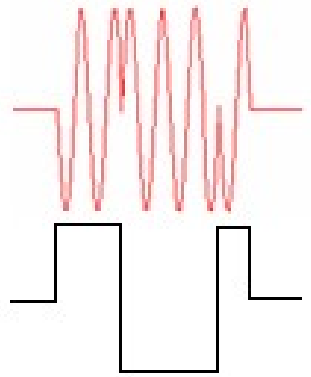
f_0=4.0MHz,
B=0.5MHz



• 2°order Butterworth band pass

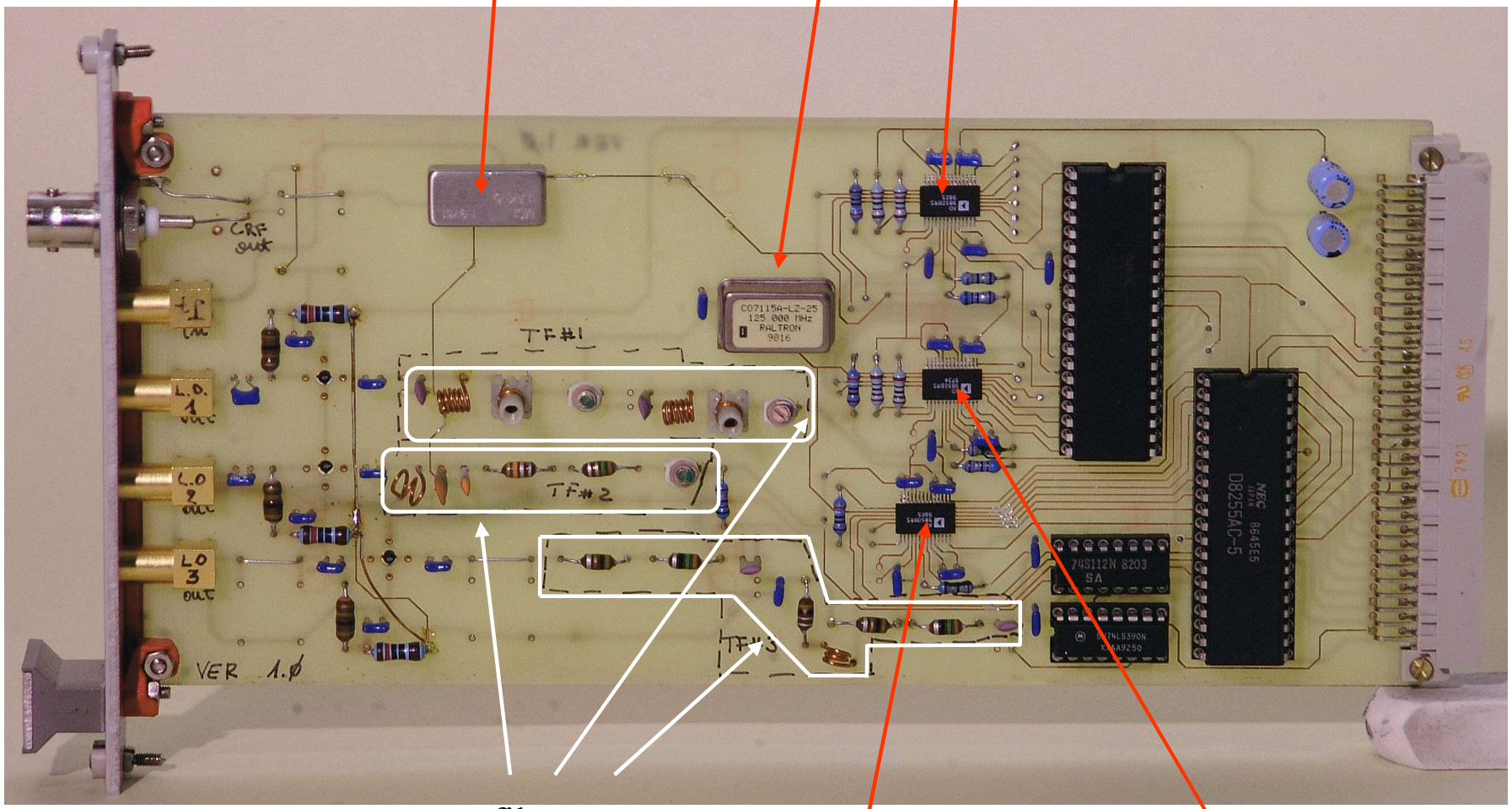
- Through mixer used as a multiplier we get coded pulses (codes coming from another board)



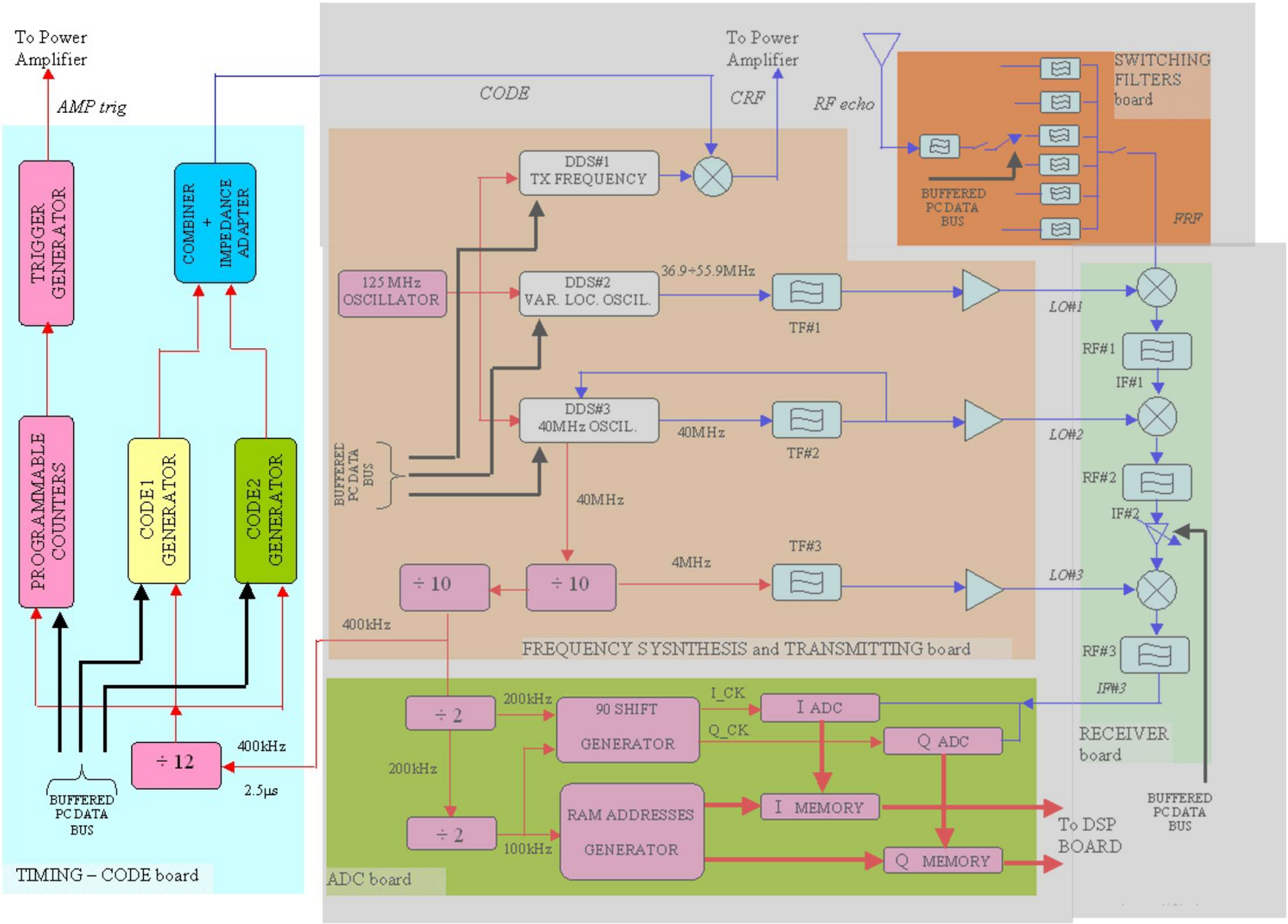


- When the code is applied at LO input automatically a coded burst of energy is created.

- Another important function of the synthesis card is the generation of a time base for the complete design.
- Starting from the LO#3 (4MHz) a 400kHz square wave is generated being the common time for the digital function.


mixer
Clock
125MHz
DDS#1
filters
DDS#3
DDS#2

TIMING AND CODE


To Power Amplifier
AMP trig
TRIGGER GENERATOR
COMBINER + IMPEDANCE ADAPTER
PROGRAMMABLE COUNTERS
CODE1 GENERATOR
CODE2 GENERATOR
÷ 12
400kHz
2.5µs
BUFFERED PC DATA BUS
TIMING – CODE board
CODE
To Power Amplifier
CRF
RF echo
SWITCHING FILTERS board
BUFFERED PC DATA BUS
FRF
DDS#1 TX FREQUENCY
125 MHz OSCILLATOR
DDS#2 VAR. LOC. OSCIL.
36.9÷55.9MHz
TF#1
LO#1
RF#1
IF#1
DDS#3 40MHz OSCIL.
40MHz
TF#2
LO#2
RF#2
IF#2
BUFFERED PC DATA BUS
40MHz
TF#3
4MHz
LO#3
RF#3
IF#3
÷ 10
÷ 10
400kHz
FREQUENCY SYSNTHESIS and TRANSMITTING board
RECEIVER board
÷ 2
200kHz
90 SHIFT GENERATOR
I_CK
Q_CK
I ADC
Q ADC
200kHz
÷ 2
100kHz
RAM ADDRESSES GENERATOR
I MEMORY
Q MEMORY
To DSP BOARD
BUFFERED PC DATA BUS
ADC board

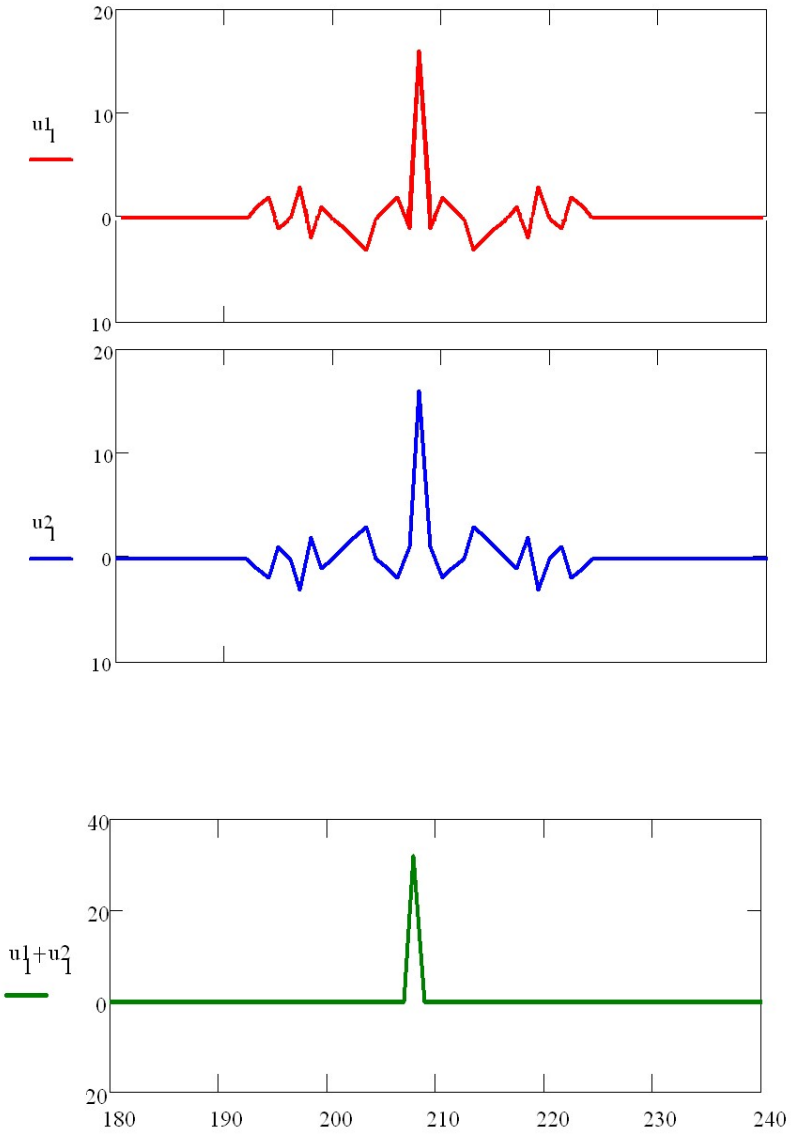
- Tasks: creates the codes to modulate the sine wave, creates the timing for the whole system.

- The code gives us the possibility to benefit of the pulse compression technique.

- Due to the ionospheric condition we can afford the complementary codes technique. This means that 2 codes are used with proper features to eliminate the side lobes "noise".

- Blue and red are the correlation peaks created by 2 complementary codes.

- The green signal is correlation result of summation: it exhibits no side lobes and an enhanced peak.

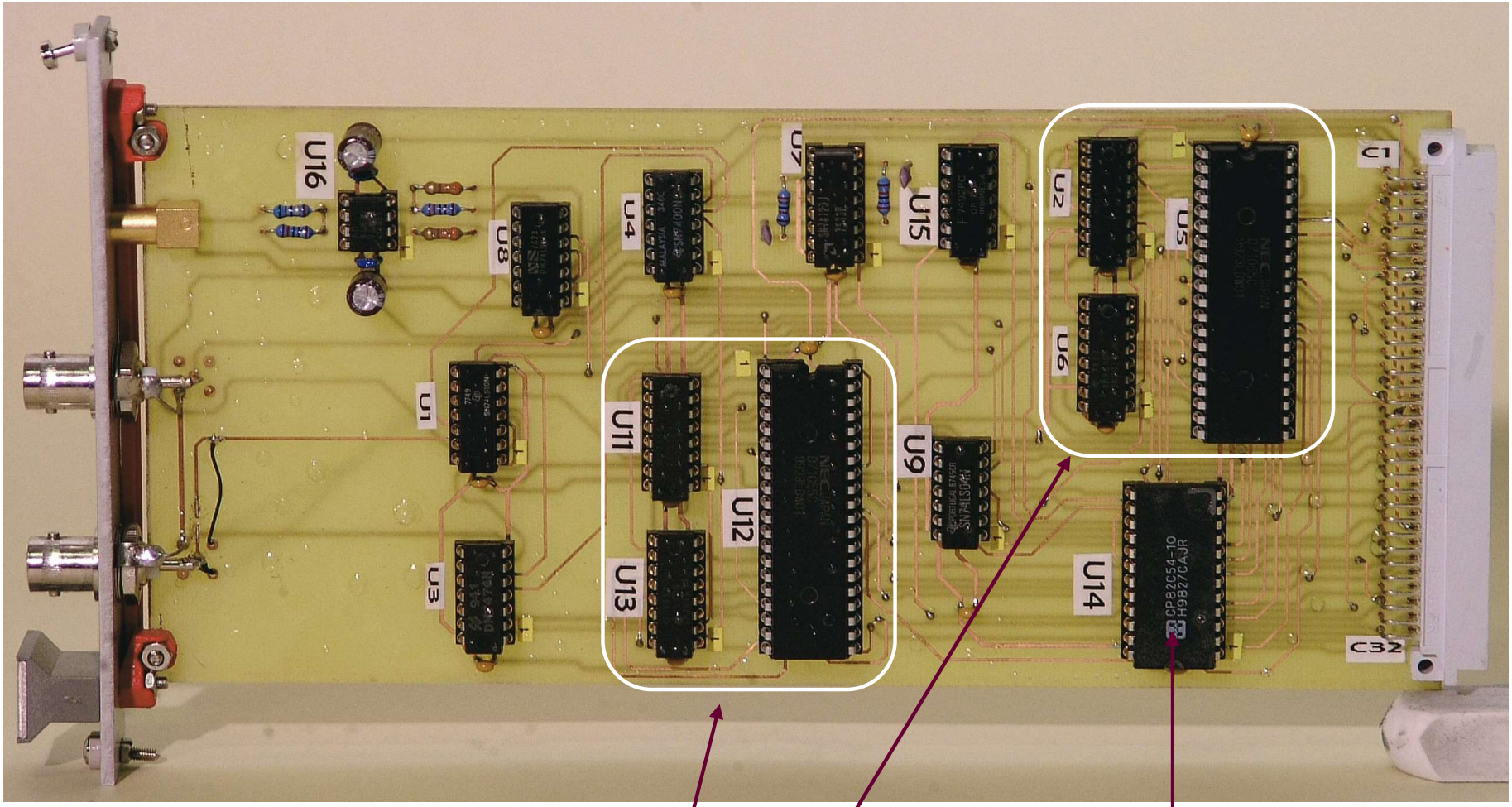


- There is a common clock so every time interval or delay is a multiple of the clock period (2.5μs). This guarantees that all signal in AIS are phase locked.

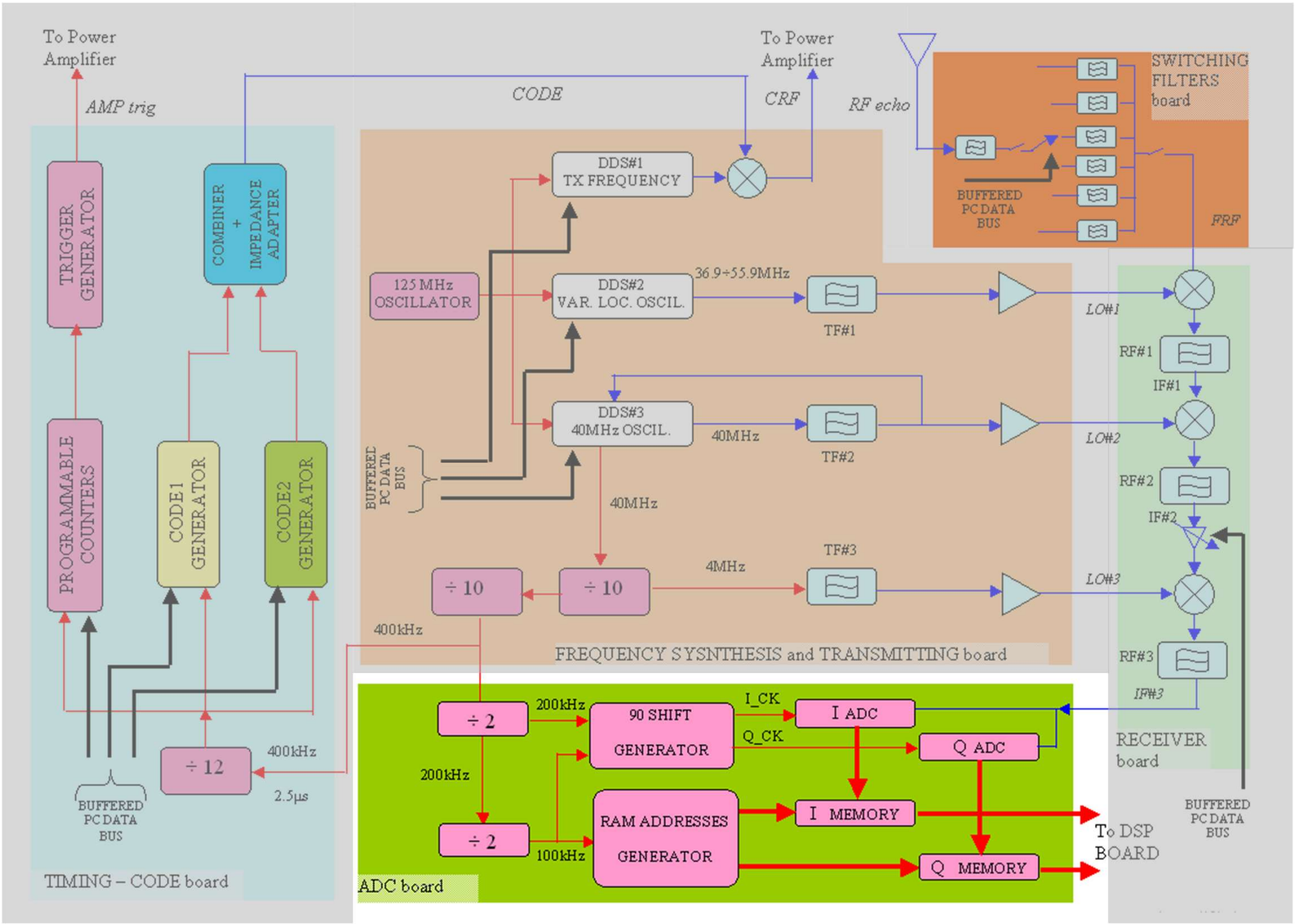
- We also used programmable counter to create well definite delays.

- We can program: pulse repetition frequency, distance between the codes, the codes themselves, number of bit representing the code and the pulse width.

- A special version of "Timing & code" board constitutes the "Echo card" in which an echo coming from a programmable fixed height is simulated.
- To create a ionospheric echo simply we create delayed codes to be used in transmitting card.

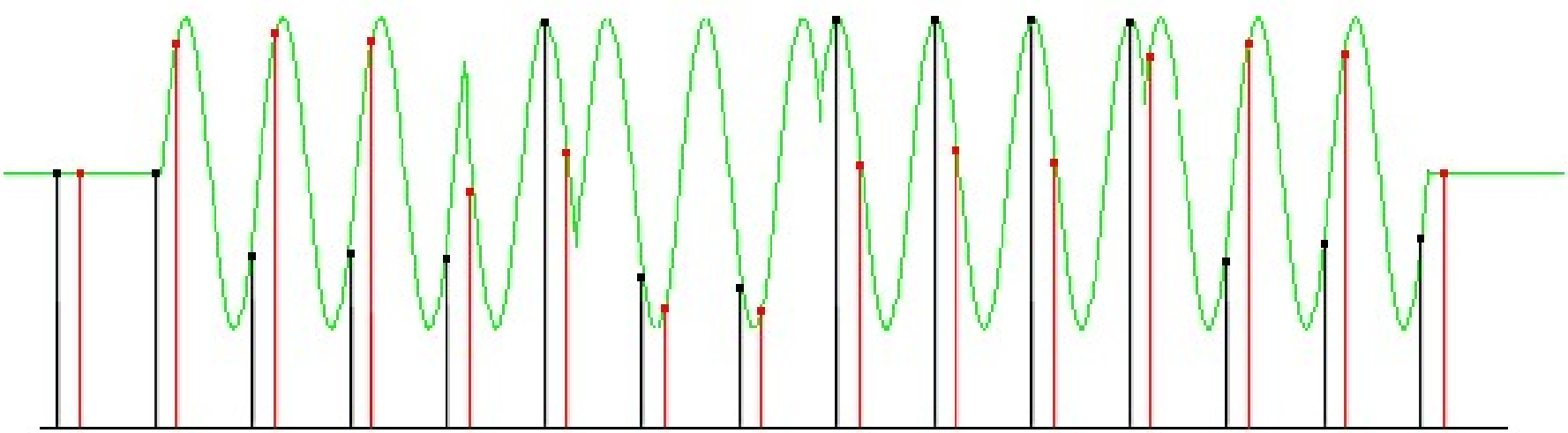

U16
U8
U4
U7
U15
U2
U5
U6
U1
U11
U12
U13
U9
U14
U3
J1
C32
Codes generators
Programmable counter

ADC


To Power Amplifier
AMP trig
CODE
To Power Amplifier
CRF
RF echo
SWITCHING FILTERS board
BUFFERED PC DATA BUS
FRF
TRIGGER GENERATOR
COMBINER + IMPEDANCE ADAPTER
DDS#1 TX FREQUENCY
125 MHz OSCILLATOR
DDS#2 VAR. LOC. OSCIL.
36.9÷55.9MHz
TF#1
LO#1
RF#1
IF#1
PROGRAMMABLE COUNTERS
CODE1 GENERATOR
CODE2 GENERATOR
BUFFERED PC DATA BUS
DDS#3 40MHz OSCIL.
40MHz
TF#2
LO#2
RF#2
IF#2
40MHz
TF#3
4MHz
÷ 10
÷ 10
LO#3
400kHz
FREQUENCY SYSNTHESIS and TRANSMITTING board
RF#3
IF#3
÷ 2
200kHz
90 SHIFT GENERATOR
I_CK
I ADC
Q_CK
Q ADC
RECEIVER board
200kHz
÷ 2
100kHz
RAM ADDRESSES GENERATOR
I MEMORY
Q MEMORY
To DSP BOARD
BUFFERED PC DATA BUS
÷ 12
400kHz
2.5µs
BUFFERED PC DATA BUS
TIMING – CODE board
ADC board

- Task: digitize the 100kHz IF to reconstruct the codes;
  store data in temporary memories to be read by DSP card.

- The basic idea is to sample signal in quadrature using 2 digitizers synchronously with the input signal obtaining the base band.
- The "I" ADC gives bytes representing "I" values (black) stored in the "I" memory, the "Q" ADC driven 90 degree phase shifted generates "Q" bytes (red) stored in the "Q" memory.



•I samples and Q samples satisfy the following relationship:

$$A_k = \sqrt{(I_k^{\,2} + Q_k^{\,2})}$$

the variation of amplitude are the code.

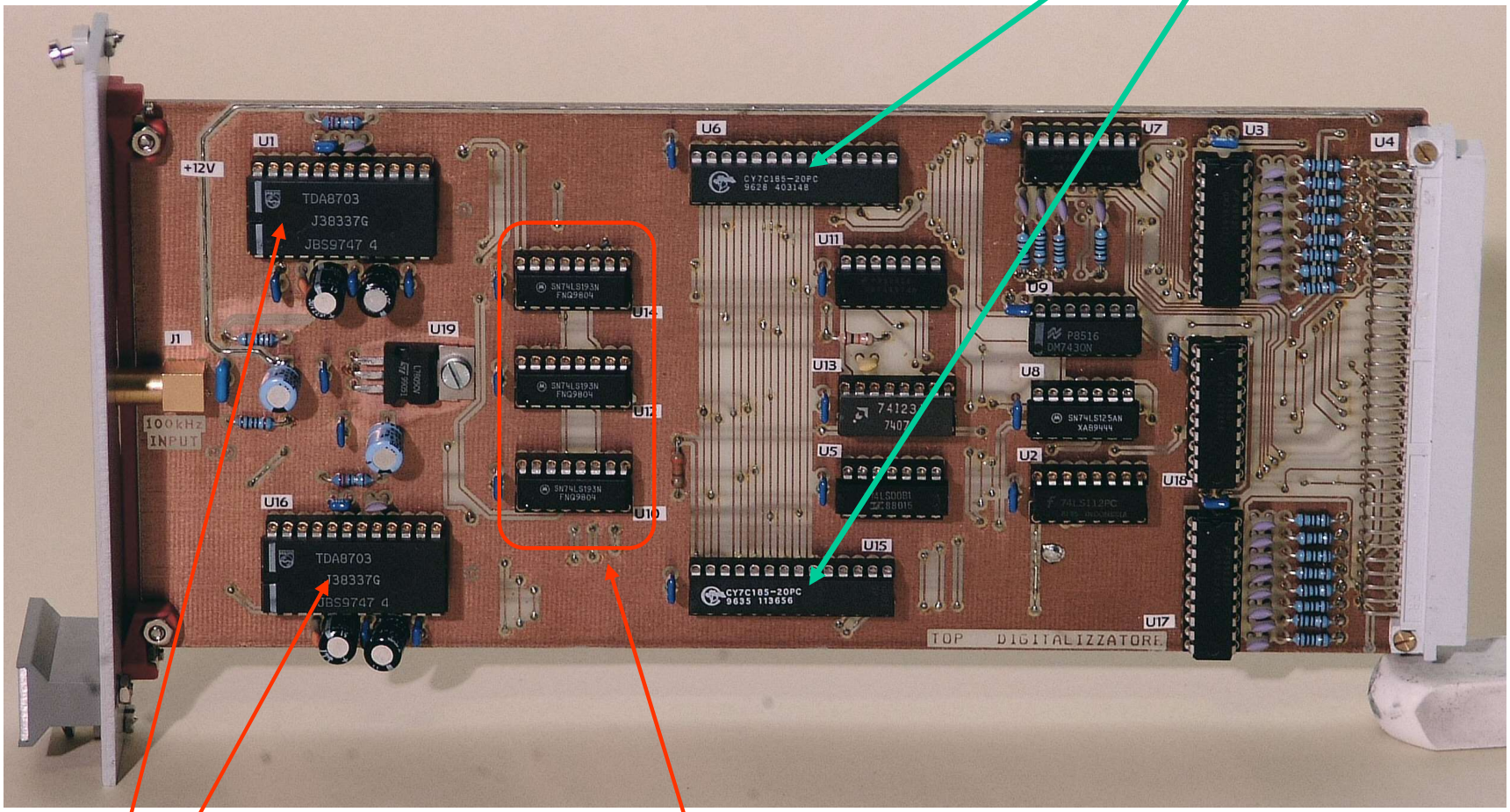
•The listening time takes approximately 5ms in which we sample 512 I values and 512 Q values stored in the two memories.

•The complete process is divided in 2 phases:

- During the listening time sampling phase is active in which last IF is digitized at 100kHz (10µs) writing data in local on board memories;
- DSP card reads the local memory importing data to analyze the echo.

•The memory addresses are generated by counters clocked synchronously with 100kHz.

On board memory



ADCs

Address generator

INTERFACE

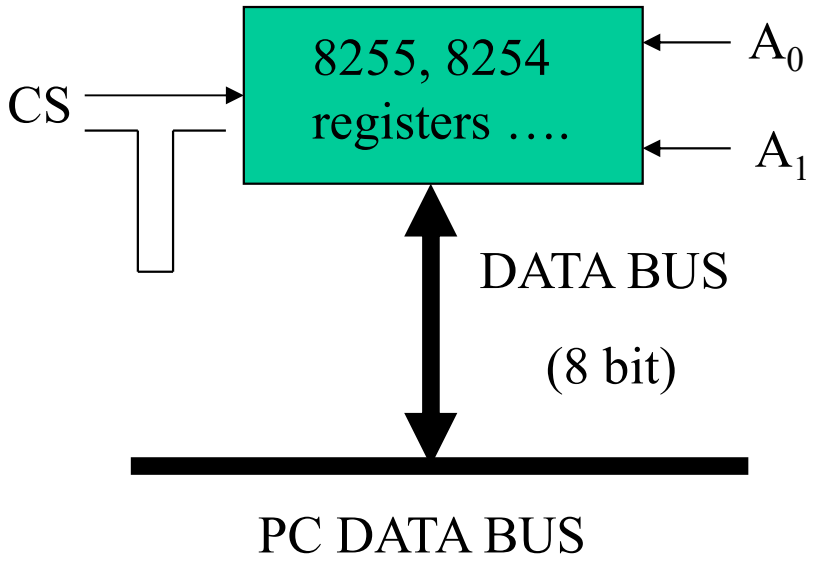
- It is a XT based bus whose tasks are:
  - -to decode addresses coming from PC (through the C language) and create proper signals to drive ionosonde's devices;
  - -to be a bridge between ADC board and DSP board (single cable communication);
  - -to allow data bus of ionosonde communicating with PC data bus.

- As every devices in PC it has an address or a group of addresses not conflicting with other peripherals.
- Valid addresses are from 300_{hex} to $31F_{hex}$ (prototype card)

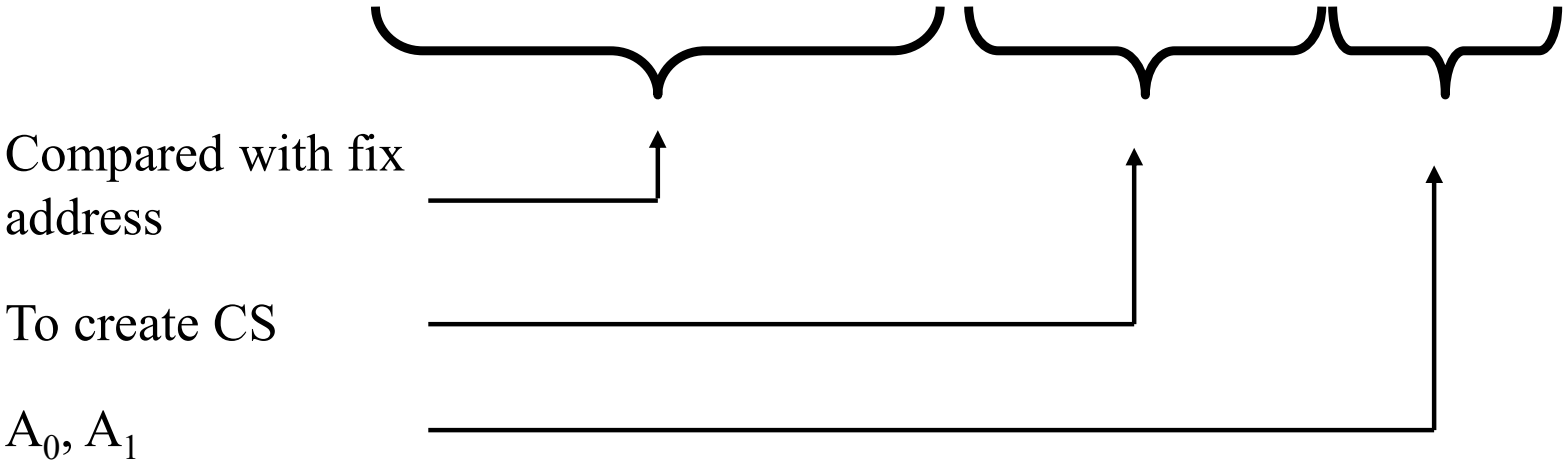
- Addresses' decoding depends on how we are going to drive our devices.
- A generic Intel device (PPI, counters) requires at least 3 control signals:

  1 Chip select (CS, active low)

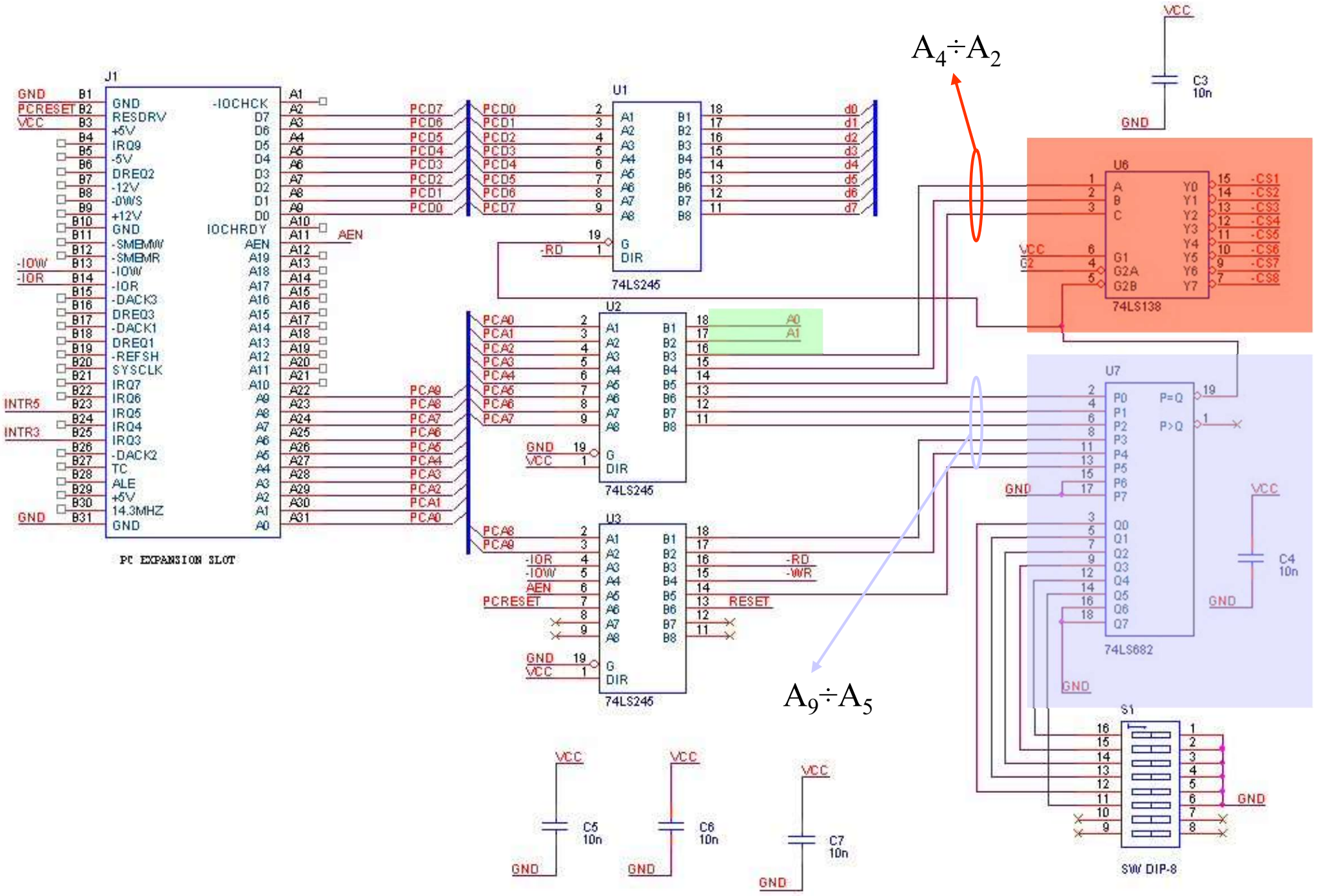
  2 address lines (A_0, A_1 for internal registers control)

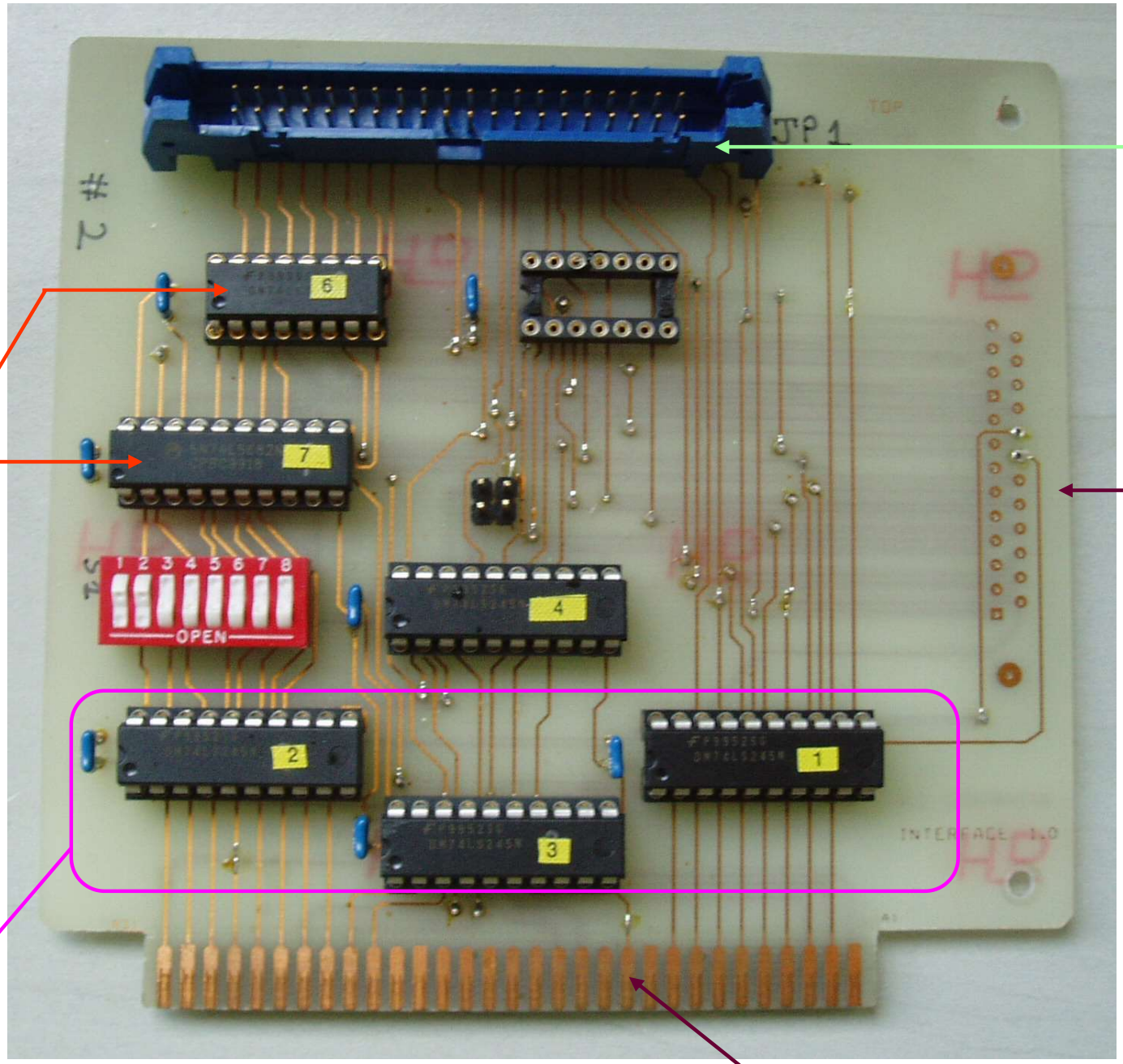


	A9	A8	A7	A6	A5	A4	A3	A2	A1	A0
300hex	1	1	0	0	0	0	0	0	0	0
31Fhex	1	1	0	0	0	1	1	1	0	0



- Different devices correspond to different CS number. We have 8 independent devices, but 32 independent addresses.
- Typical instruction: **outport(0x30C , 20) ;**


$A_4 \div A_2$
$A_9 \div A_5$
J1
PC EXPANSION SLOT
U1
74LS245
U2
74LS245
U3
74LS245
U6
74LS138
U7
74LS682
S1
SW DIP-8
C3 10n
C4 10n
C5 10n
C6 10n
C7 10n
VCC
GND
-RD
-WR
RESET
AEN
A0
A1
INTR5
INTR3
-IOW
-IOR
PCRESET
-CS1
-CS2
-CS3
-CS4
-CS5
-CS6
-CS7
-CS8


To DSP card
To main unit
comparators
buffers
ISA bus
JP 1
OPEN

DSP

- It is an ISA Bus board based upon the Texas Instruments TMS320C5x series fixed point DSP delivering up to 40 MIPS performance.
- Task: to perform the analysis of the received echo to detect the codes and to find the height of the layer.
- This task is accomplished with a program, running inside the board, loaded at the beginning of the sounding, that is independent of the main sounding software.
- The complete process from echo to a dot on the screen goes through some phases:
  - at the end of listening time, ADC card calls DSP that starts operation;
  - DSP reads 512 samples (16 bit words), and start the analysis (code recognition, FFT, filtering, correlation, integration, codes summation);
  - when the programmed number of integration is reached DSP calls PC that will complete the analysis (IFFT, thresholding) creating the ionogram dot by dot.
- DSP dialogues with ADC directly through a connection with interface board so to use one cable only.

Overview

The Model 5000 is an ISA Bus board based upon the Texas Instruments TMS320C5x series fixed point DSP delivering up to 40 MIPS performance. Intelligent data acquisition is provided for 8 input channels at an aggregate sampling rate of 500 KSPS. Two analog outputs are provided, as well as buffered digital I/O expansion.

Key Features

- TI TMS320C5x DSP at 80 MHz
- 64K Words Program RAM
- 128K Words Data RAM
- 500 KSPS 12 bit A/D Converter
- 8 input multiplexer
- Dual 12 bit D/A Converter
- Digital I/O
- DSP Serial I/O
- ISA Bus Interface

Software

- Assembler and Debugger
- Link Package
- Real time Display
- Data Acquisition Manager
- Application examples

All these efforts to create.....



And now.....AUTOSCALA!!!!